UNIT – I : SOIL EXPLORATION

Foundation Engineering (23A01604c) · Handwritten Notes

1.1 Introduction

- Every structure ends up on the ground, so the **soil must carry the load safely.**
- Soil exploration (= site investigation / sub-surface exploration) is the process of finding the **soil profile, soil properties and ground water table** at a site.
- It is the **first step** of foundation design – cheapest part of the job (0.5 – 1 % of project cost) but a wrong investigation can cost many times more.

> **Definition:** Soil exploration is the planned collection of information on the sub-soil – its stratification, engineering properties, GWT and rock level – by field and laboratory methods so that a safe and economical foundation can be designed.

1.2 Need for Soil Exploration

1. To know the **nature of soil deposits** – sequence, thickness, extent of the strata.
2. To find the **depth of ground water table** (GWT) and its fluctuation.
3. To determine **engineering properties** – shear strength (c, φ), compressibility (C_c, m_v), permeability, density.
4. To select the **type and depth of foundation** (shallow / pile / well / raft).
5. To estimate **safe bearing capacity** and probable **settlement.**
6. To predict & prevent problems – swelling, collapse, liquefaction, scour, aggressive soil/water (sulphates, chlorides).
7. To select **construction method** – dewatering, shoring, bracing of excavations.
8. To investigate **failure / distress** of structures, slopes, dams; to locate borrow areas.

1.3 Stages of Investigation

- **Reconnaissance** – desk study, geological maps, site walk-over, existing records.
- **Preliminary exploration** – few borings, to find general soil profile, rock depth, GWT.
- **Detailed exploration** – close spaced borings, sampling and in-situ tests.

1.4 Planning of the Exploration Programme

The programme depends on the type of structure, its importance, loads and the soil variability. Steps:

1. Collect **data on the proposed structure** – type, size, column loads, tolerable settlement.
2. Collect **data of the site** – topography, geology, nearby structures, past land use (fill, mine, tank).
3. Fix **number, location, depth** of boreholes / pits.
4. Choose **method of boring and sampling**; decide in-situ tests (SPT, CPT, vane, PLT).
5. Decide **laboratory tests** to be done on samples.
6. Collect results → prepare **borelogs, soil profile, report.**

(a) Depth of exploration

- Boring must go to the depth where the **stress increase $\Delta\sigma$ is only ≈ 10 % of the foundation pressure q** (i.e. the "significant depth"), or until hard rock/stratum.
- IS 1892 : at least **1.5 × width** of foundation (isolated); deeper for strip/raft.

Foundation / structure	Minimum depth of boring
Isolated square footing (width B)	≈ 1.5 B – 2 B below foundation level
Strip footing (long)	≈ 3 B – 4 B
Raft (width B)	≈ 1.5 B (and ≥ 6 m) below raft
Pile foundation	≥ 5 m below pile tip / 1.5 × group width
Highway cutting / embankment	≈ 1 – 2 × height; ≥ 3 m below subgrade

(b) Spacing of boreholes

Project	Spacing (m)
Multi-storey buildings	10 – 30
One-storey industrial plants	20 – 60
Highways	150 – 300
Earth dams	15 – 60

- Minimum **3 boreholes** not in a straight line (to get dip of strata); more when soil is erratic.

(c) Typical layout of boreholes

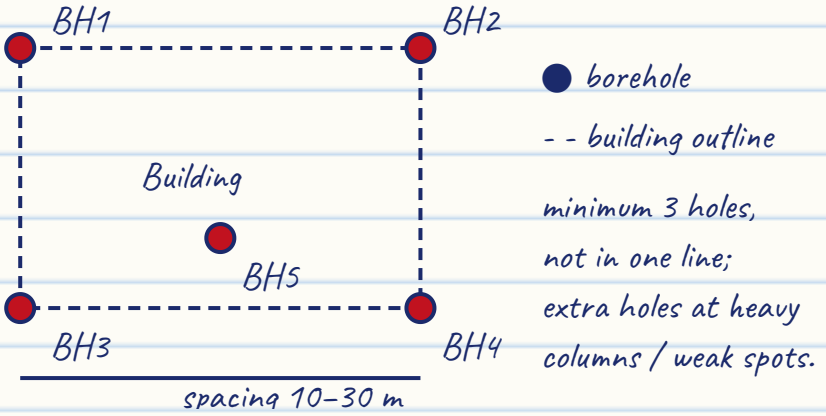


Fig 1.1 Borehole layout for a building

I.5 Methods of Soil Exploration

Classification:

- **Direct methods** – open excavations: test pits, trenches, shafts, drifts.
- **Semi-direct methods** – borings (auger, wash, percussion, rotary) with sampling.
- **Indirect methods** – geophysical (seismic, electrical resistivity), sounding/penetration (SPT, CPT, DCPT), vane.

A. Open excavations (direct)

- **Test pits** – 1.2 m × 1.2 m (min) or larger; depth up to 3 – 4 m (above GWT).
- **Trenches** – long narrow cuts for strata continuity (e.g. along a pipeline, fault).
- **Shafts / drifts** – large; used for dam sites in rock.

Advantages:

- Soil is visible **in-situ** → exact stratification, fissures, lenses seen; best undisturbed (block) samples; in-situ tests like PLT possible.

Limitations:

- Costly and slow beyond 3 – 5 m; needs shoring/dewatering below GWT; unsafe in loose sand/soft clay.

B. Boring methods

Used when the depth is large. Hole diameter 100 – 200 mm (IS 1892).

I. Auger boring

- **Hand auger** (post-hole / helical) – 100 – 200 mm dia; depth ≤ 5 – 6 m; for soft to stiff cohesive soils above GWT. Soil is brought up in the auger – disturbed sample.
- **Power (flight) auger** – continuous flights rotated by machine; depth up to 30+ m in cohesive soil; hollow-stem type allows SPT/sampling through the stem.

- Not suitable in gravel, boulders, below GWT in loose sand (hole collapses).

2. Wash boring

- Casing (hole lining) is driven; a **chopping bit** on hollow drill rod loosens soil while **water is jetted** through the rod.
- Water + cuttings rise in annulus to a settling tank; cuttings show the soil change only roughly.
- Suitable for sand, silt and soft clay; not for gravel / boulders / hard clay. Cheap, quick; SPT can be done at intervals.

3. Percussion (cable-tool / churn) drilling

- A heavy chisel bit is repeatedly **lifted and dropped** on a cable; soil/rock is broken, mixed with water to slurry and removed by a bailer/sand pump.
- Good for hard clay, gravel, boulders, soft rock; casing advanced simultaneously. Slow. Samples are crushed → disturbed.

4. Rotary drilling

- A **rotating bit** (drag/ roller/ diamond) on drill rods cuts the soil or rock; drilling fluid (bentonite mud / water) is pumped down, carries cuttings up and keeps the hole open.
- Fast, can go very deep; **core barrel** with diamond bit gives **rock cores** (NX, BX, AX sizes).
- Used for rock, hard soil, deep investigations (dam, bridge sites); cost is high.

Comparison of boring methods

Method	Soil type	Depth	Sample
Hand auger	soft/stiff clay, above GWT	≤ 6 m	disturbed
Power auger	clays, silts, dense sand	≤ 30 m	disturbed
Wash boring	sand, silt, soft clay	≤ 30 – 60 m	cuttings
Percussion	all soils incl. gravel, soft rock	deep	disturbed
Rotary	all soils & rock	very deep	cores

C. Geophysical (indirect) methods

Measure physical properties (velocity, resistivity) from the surface – **quick, cheap, no sample**; results must be calibrated with a few borings.

(i) Seismic refraction method

- Shock wave created by hammer/explosive at surface; **geophones** record first arrival times at different distances.
- Time–distance plot gives wave velocities V_1, V_2 of layers and the thickness of top layer:

$$H = (x_c / 2) \cdot \sqrt{[(V_2 - V_1)/(V_2 + V_1)]}$$

x_c = critical distance (break in the T–x graph). Needs $V_2 > V_1$.

(ii) Electrical resistivity method (Wenner)

- Four equally spaced electrodes (spacing a) – current I through outer two, potential V across inner two.

$$\rho = 2\pi a V / I$$

- Resistivity changes with moisture, salt content & soil type: **sharp drop at GWT**; high for rock/dry gravel, low for clay. Used for location of GWT, rock, buried channels, corrosion studies.

1.6 Soil Sampling

Purpose: to get soil specimens for visual classification and lab tests (grain size, Atterberg limits, c, φ, consolidation, permeability).

Classification of samples

- **Disturbed – non-representative:** soil mixed with other strata / washed fines (e.g. wash cuttings). Only for rough identification.
- **Disturbed – representative:** structure destroyed but all constituents and water content retained (split-spoon, auger samples, kept in air-tight jar). Used for grain size, Atterberg limits, specific gravity, compaction.
- **Undisturbed:** structure, density, water content unchanged as far as possible (thin-walled tube, block samples). Needed for **shear strength, consolidation, permeability** tests on clays.

No sample is truly "undisturbed" – sampling disturbance is only minimised.

Sampling methods

1. **Auger / test-pit samples** – disturbed.

2. **Chunk (block) samples** – cut by hand from pit, coated with paraffin wax, wrapped & boxed – best quality, undisturbed.
3. **Split-spoon (SPT) sampler** – disturbed representative; 35 mm ID.
4. **Open drive (thin-wall Shelby tube)** sampler – undisturbed in soft–medium clays; tube pushed at a steady rate.
5. **Stationary piston sampler** (Osterberg / Hvorslev) – piston keeps soft clay from entering before the tube is pushed; better recovery in soft clays.
6. **Rotary (Denison, Pitcher) samplers** – double-tube with a rotating outer cutting shoe, for stiff / hard clay and soft rock.
7. **Sand sampling** – hard to sample; core-catcher / freezing technique, or in-situ density by sand-replacement & core-cutter methods.

Handling of samples

- Seal both ends with wax, label (project, BH no., depth, date), keep upright, protect from sun, vibration and drying; test early.

1.7 Design Features of a Sampler

Sample disturbance depends on the **geometry of the cutting edge**. Let

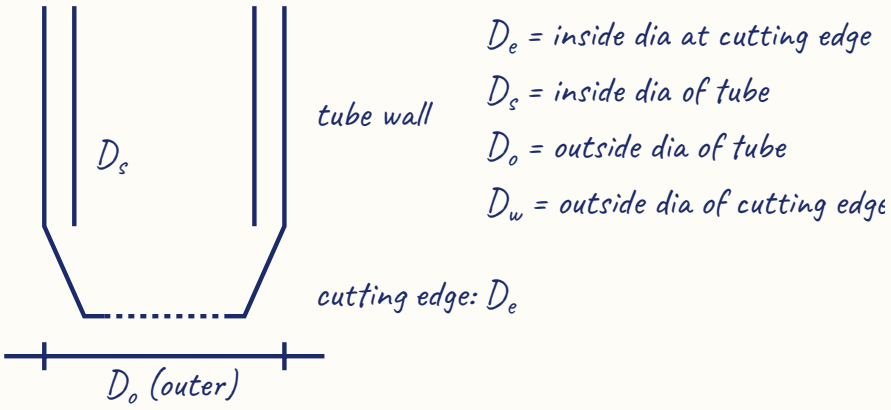


Fig 1.3 Thin-walled tube sampler – shoe details

$$\text{Area ratio } A_r = (D_w^2 - D_e^2)/D_e^2 \times 100\ \%$$
$$\text{Inside clearance } C_i = (D_s - D_e)/D_e \times 100\ \%$$
$$\text{Outside clearance } C_o = (D_w - D_o)/D_o \times 100\ \%$$
$$\text{Recovery ratio } L_r = L / H \ (L = \text{sample length}, H = \text{penetration})$$

- A_r = volume of soil displaced ÷ volume of sample. For undisturbed sample $A_r \leq 10\ \%$ (thin-wall; IS 2132 $\leq 20\ \%$ for thick-wall tubes).

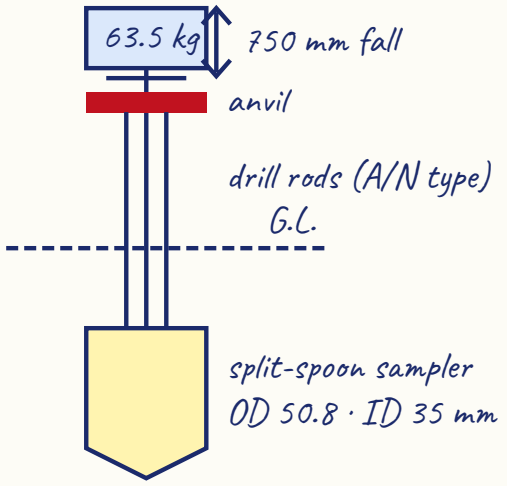
- C_i = 0.5 – 3 % : lets sample expand slightly, reduces wall friction (small clearance for clays, up to 3 % for long tubes).
- C_o = 0 – 2 % : eases withdrawal of the tube.
- **Recovery ratio** should be ≈ 96 – 98 % (≥ 100 % means swelling, < 100 % shortening).
- Cutting edge angle small (≈ 10°–12° for thin tubes), tube length ≈ 10 – 20 × D, a **vent / ball valve** at the head to release air and hold the sample (suction).

Split-spoon sampler: A_r ≈ 110 % (50.8 & 35 mm) – hence **disturbed**.

Field Tests

1.8 Standard Penetration Test (SPT) – IS 2131

Most widely used in-situ test, especially in **cohesionless soils** where undisturbed sampling is difficult.



① seating drive 150 mm
② test drive 2 × 150 mm
→ blows counted = N
③ stop if N > 50 (refusal)

Fig 1.4 SPT assembly

Procedure

1. Bore hole cleaned to required depth (150 – 200 mm dia); casing used if sides cave.
2. Split-spoon (length 450 mm) lowered at the bottom with drill rods.
3. Hammer of **63.5 kg (140 lb)** dropped freely through **750 mm** on the anvil.
4. Sampler driven **450 mm** in 3 steps of 150 mm: blows for the **first 150 mm are neglected** (seating).
5. N-value = blows for the last 300 mm (2nd + 3rd drive). If 50 blows do not give 150 mm → **refusal**, record penetration.
6. Sampler is withdrawn, split opened and soil is taken as a representative sample.

7. Test done every **1.0 – 1.5 m** depth or at change of stratum.

N is the number of blows per 300 mm penetration (blows/ft in the old units).

SPT corrections

(1) Dilatancy (water-table) correction – for fine / silty sand below GWT (Terzaghi & Peck), when observed $N > 15$:

$$N_c = 15 + \tfrac{1}{2}(N - 15)$$

(excess pore pressure in dense fine sand makes N too high; no correction if $N \leq 15$ or in coarse sand/gravel).

(2) Overburden pressure correction – N increases with depth, so reduce to a standard σ'_o:

$$N' = C_N \cdot N \text{ with } C_N = 0.77 \log_{10}(2000 / \sigma'_o) \; [\sigma'_o \text{ in kN/m}^2,\ \geq 25]$$

$$\text{(Liao–Whitman: } C_N = \sqrt{(100/\sigma'_o)}\text{)}$$

(3) Energy / other corrections (modern practice, N_{60}):

$$N_{60} = N \cdot \eta_H \eta_B \eta_S \eta_R / 0.60$$

- η_H hammer efficiency (safety hammer 0.6–0.75, donut 0.45–0.6, automatic 0.8–1.0)
- η_B borehole dia (1.0 for 65–115 mm, 1.05 for 150 mm, 1.15 for 200 mm)
- η_S sampler (1.0 with liner, 1.2 without liner)
- η_R rod length (0.75 for < 4 m, 0.85 for 4–6 m, 0.95 for 6–10 m, 1.0 for > 10 m)

Order of applying (IS): first dilatancy correction, then overburden correction (apply dilatancy only when $N > 15$ in saturated fine sand).

N-value correlations (Terzaghi & Peck)

Sand N	Compactness	Dr %	φ
0–4	very loose	<15	<28°
4–10	loose	15–35	28–30°
10–30	medium	35–65	30–36°
30–50	dense	65–85	36–41°
>50	very dense	>85	>41°

Clay (N → consistency, q_u kN/m²): <2 very soft (<25); 2–4 soft (25–50); 4–8 medium (50–100); 8–15 stiff (100–200); 15–30 very stiff (200–400); >30 hard (>400).

Worked Example 1 – SPT corrections

SPT in fine saturated sand at 6 m depth gave $N = 32$. $\gamma_{sat} = 19$ kN/m³, GWT at 2 m below GL, $\gamma_w = 10$ kN/m³. Find the corrected N.

- $\sigma'_o = 2(19) + 4(19 - 10) = 38 + 36 =$ **74 kN/m²** (taking γ above GWT also 19)
- Dilatancy: $N_c = 15 + 0.5(32 - 15) = 15 + 8.5 =$ **23.5**
- $C_N = 0.77 \log(2000/74) = 0.77 \times 1.432 =$ **1.10**
- $N' = 1.10 \times 23.5 =$ **25.9 ≈ 26** → medium dense sand ($\varphi \approx 33°$)

Uses of SPT results: relative density, φ, bearing capacity (Meyerhof, Peck), settlement of footings, pile capacity, liquefaction check, c_u of clays ($c_u \approx 6\,N$ kN/m²).

Merits / limitations of SPT

- \+ Simple, cheap, gives sample, done in any borehole, wide database.
- – Poor repeatability (hammer type, operator, rod energy); disturbed sample; not reliable in clays, gravels (cobble gives false high N).

1.9 Dynamic Cone Penetration Test (DCPT) – IS 4968 (Part 1 & 2)

- A **cone (60° apex)** is driven by a **65 kg hammer, 750 mm fall**; no sample; used mainly in sands/gravels.
- **Type A (50 mm cone):** blows per **100 mm** penetration = N_{cd}. Used up to ~ 8 m with no casing.
- **Type B (65 mm cone) + bentonite slurry:** blows per **300 mm** = N_{cd}; slurry reduces rod friction, deep up to 20 m+.
- Continuous record of penetration resistance with depth; correlated with SPT: $N_{cd} \approx N$ (approx.) – found for a site by calibration.

1.10 Static Cone Penetration Test (SCPT / CPT) – IS 4968 (Part 3)

Also called **Dutch cone test.** A cone is pushed (not driven) into the ground at a constant rate of **20 mm/s** and the resistance is measured.

- **Cone resistance** $q_c = Q_c / A_c$ ($A_c = 10$ cm²)
- **Sleeve (skin) friction** $f_s = Q_s / A_s$ ($A_s = 150$ cm²)
- **Friction ratio** $R_f = f_s / q_c \times 100\,\%$ → identifies soil: sand $R_f < 1\,\%$, clay 3 – 5 %.

- Electric cone / **piezocone (CPTu)** measures q_c, f_s and pore pressure u continuously.

Applications / correlations

- Sands: $q_c \approx 400\ N$ (kN/m²) (N = SPT); φ from q_c charts; bearing capacity and settlement (Schmertmann).
- Clays: $c_u = (q_c - \sigma_{vo}) / N_k$, $N_k = 15 - 20$.
- Pile capacity: end bearing ≈ q_c, skin friction ≈ f_s (Dutch method).

SPT vs CPT: SPT is dynamic, gives a disturbed sample, discrete readings; CPT is static, no sample, continuous record, not for gravel.

I.11 Vane Shear Test – IS 4434

An in-situ test to get the **undrained shear strength c_u of soft, saturated clays** and their **sensitivity** (no sampling disturbance).

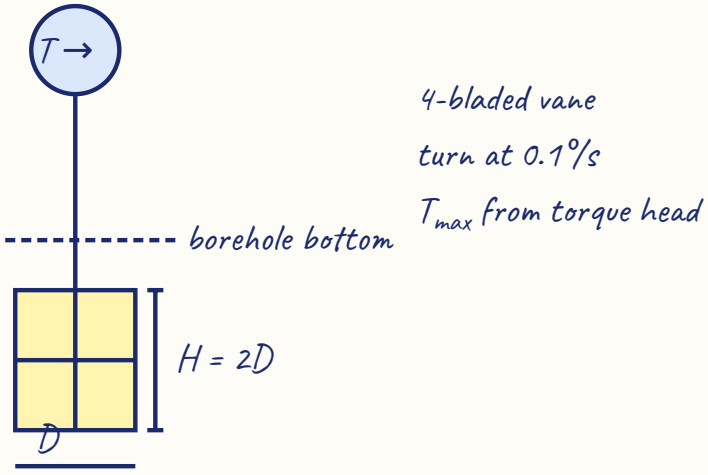


Fig 1.7 Vane shear apparatus

- Vane (two orthogonal blades, typ. D = 75 mm, H = 150 mm, so H/D = 2) is pushed to the test depth, then **torque** applied at ≈ 0.1°/s till failure of the cylinder of soil. Peak T → c_u.
- Rotate rapidly 10 times – **remoulded** strength measured again.

Torque resisted by cylindrical surface + top & bottom circular ends:

$$T = \pi D^2 H c_u/2 + 2 \cdot (\pi D^3 c_u/12) \Rightarrow c_u = T / [\pi (D^2H/2 + D^3/6)]$$

$$H = 2D : c_u = T / (3.66\ D^3)$$

$$\text{Sensitivity } S_t = c_u(\text{undisturbed}) / c_u(\text{remoulded})$$

- Suitable for soft clay (c_u < 100 kN/m²); not in sandy/ stony or stiff clays; correct c_u for plasticity (Bjerrum μ) for embankment stability.

Example: $D = 75$ mm, $H = 150$ mm, $T = 38$ N·m: $c_u = 38 / (3.66 \times 0.075^3) = 38 / (1.544 \times 10^{-3}) =$ **24.6 kN/m²** (soft clay).

1.12 Plate Load Test (PLT) – IS 1888

A field loading test on a small rigid plate placed at foundation level to obtain the **load–settlement behaviour**, ultimate bearing capacity and the modulus of subgrade reaction.

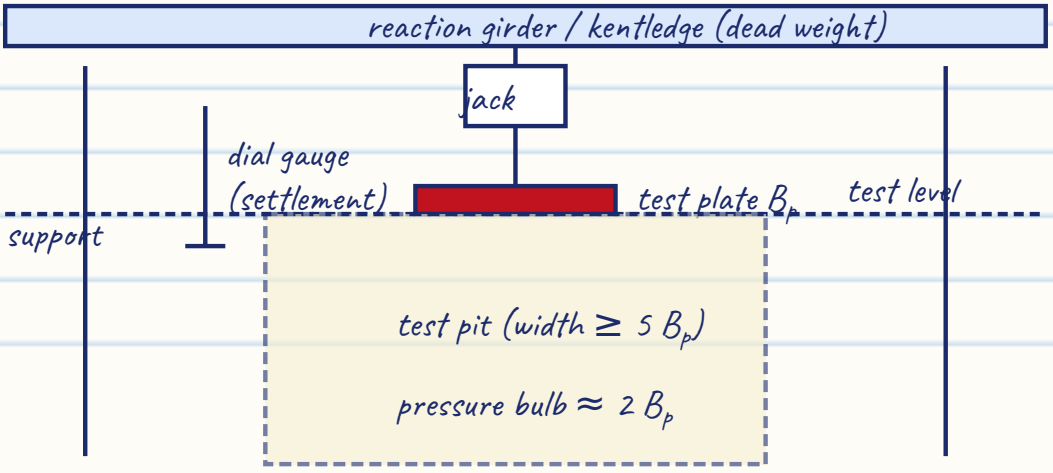


Fig 1.8 Plate load test set-up (gravity loading)

Equipment & procedure

- Plates: **mild steel, 25 mm thick, 300 – 750 mm square or circular** (300, 450, 600, 750 mm); sand bedding.
- Test pit width ≥ 5 B_p; bottom levelled at proposed foundation level; GWT lowered if above.
- Load applied by **hydraulic jack** against kentledge (or anchor / truck reaction); settlements by **2 – 4 dial gauges** (0.01 mm) on independent datum beam.
- Seating load ≈ 7 kN/m² released; then **increments = 1/5 of estimated safe load** (≈ 1/10 of ultimate).
- Each increment kept until **rate of settlement ≤ 0.02 mm/min** (or 1 h in clays); readings at 1, 2.25, 4, 6.25, 9, 16, 25… min.
- Continue to **failure** or to **1.5 – 2 × the design load** / settlement of 25 mm.

Load–settlement curve & ultimate load

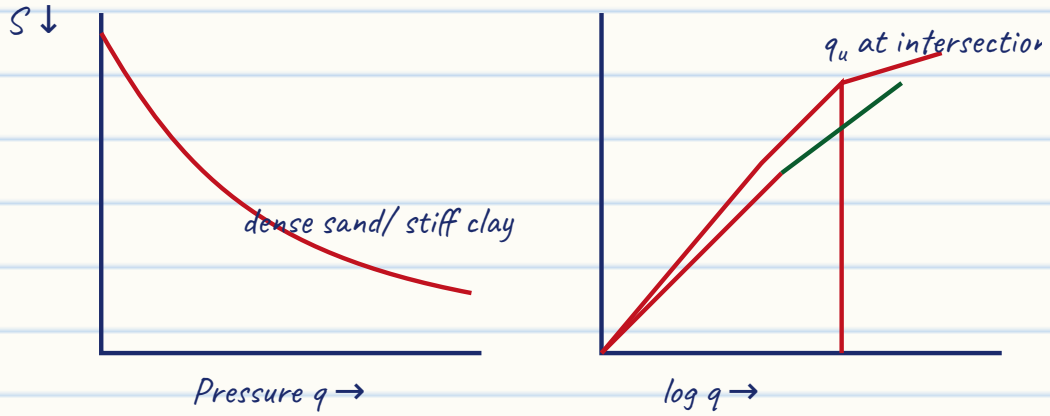


Fig 1.9 Load–settlement curves; ultimate load from log–log plot (two straight lines intersect)

- **Clear failure** (dense sand, stiff clay): peak/ break of curve = q_u.
- **No clear failure** (loose sand, soft clay): double tangent or log q – log S intersection; or load at S = 25 mm (or 10 % of B).

Size (scale) effect – Terzaghi & Peck

Footing size B_f and plate B_p:

Soil	Bearing capacity	Settlement
Clay (cohesive)	$q_{uf} = q_{up}$ (independent of size)	$S_f = S_p \cdot (B_f / B_p)$
Sand (cohesionless)	$q_{uf} = q_{up} \cdot (B_f / B_p)$	$S_f = S_p \cdot [B_f (B_p + 0.3) / (B_p (B_f + 0.3))]^2$ (B in m)

For c–φ soils, the sand rule is used on the safe side. Safe pressure = q_u / FS (2.5 – 3) with S ≤ allowable (25 mm isolated footing in sand).

Worked Example 2

PLT on sand with 0.3 m square plate: q_{up} = 180 kN/m², settlement 10 mm at design pressure. Footing 1.5 m square. Find q_{uf} and S_f.

- q_{uf} = 180 × 1.5/0.3 = **900 kN/m²**; with FS = 3 → q_{safe} = **300 kN/m²**
- S_f = 10 × [1.5 (0.3 + 0.3) / (0.3 (1.5 + 0.3))]² = 10 × (0.9/0.54)² = 10 × 2.78 = **27.8 mm** → if > 25 mm, reduce pressure

Limitations of the plate load test

- Influences only ≈ 2 B_p depth – cannot sense deeper weak layers (clay below sand) that the large footing stresses.
- Does **not** give long-term consolidation settlement of clays (short duration test).

- Size effect – results must be corrected; GWT effects; costly and time consuming.
- Test the **sub-grade modulus:** $k_s = q / S$ (kN/m³) for plate of 0.3 m; footing: $k_f = k_p (B_p/B_f)$ for clay; $k_f = k_p [(B_f+0.3)/(2B_f)]^2$ for sand.

1.13 Pressuremeter Test (Ménard)

A cylindrical probe is expanded **radially** inside a borehole; the pressure–volume change gives **in-situ stress-strain modulus and strength** of soil/soft rock.

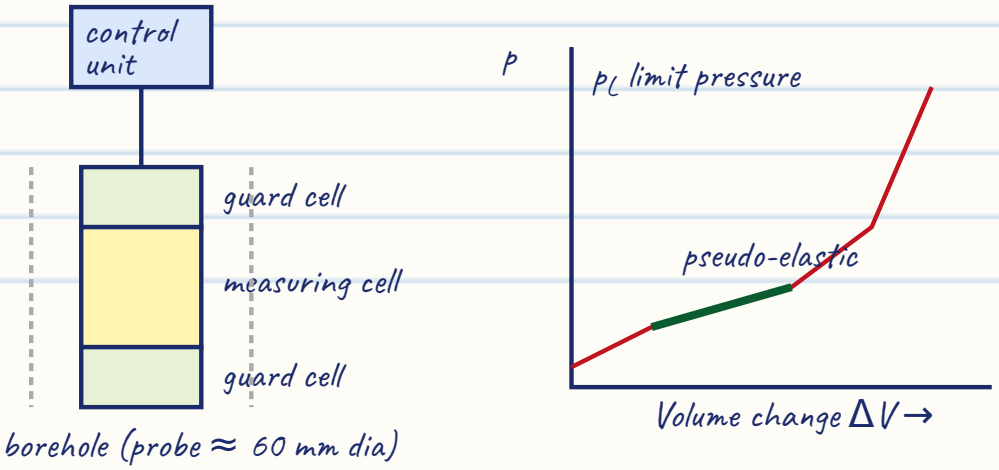


Fig 1.10 Ménard pressuremeter & pressure–volume curve

- Probe: 3 cells – central **measuring cell** (water-filled) + two **gas-filled guard cells** (keep the strain plane-cylindrical).
- Pressure raised in equal steps (each 1 min), volume change read at 15, 30, 60 s.

Results of pressuremeter

$$E_m = 2(1+\mu)(V_0 + V_m) \cdot \Delta p / \Delta V$$

- V_0 = initial volume of measuring cell, V_m = volume at middle of the straight (pseudo-elastic) portion, μ = Poisson's ratio (≈ 0.33), $\Delta p/\Delta V$ = slope of that portion.
- p_0 – at-rest horizontal pressure (start of straight part); p_f – creep pressure (end of straight part); p_L – limit pressure (volume = 2 V_0 … cavity doubles).
- Bearing capacity (Ménard): $q_u = \sigma_{v0} + k(p_L - p_0)$, k = bearing factor (0.8 – 3) depending on soil & D_f/B; safe $q = q_u/3$.

Advantages: in-situ, gives modulus + strength, works in all soils & weak rock. Limitations: borehole disturbance, costly, interpretation empirical.

1.14 Soil Investigation Report

Final document handed to the designer (IS 1892): must be **clear, complete, honest**, with data separated from opinion.

1. **Introduction** – project, client, purpose, dates, scope of work.
2. **Site description** – location, topography, geology, nearby structures, climate, drainage.
3. **Field investigation** – number, location, depth of boreholes/pits, method, in-situ tests, sampling, GWT observed.
4. **Laboratory tests** – classification, c, φ, C_c, permeability, chemical tests, rock tests.
5. **Soil profile** – borelogs, cross-sections, strata description.
6. **Analysis & recommendations** – type and depth of foundation, **safe bearing capacity**, settlement, ground improvement, dewatering, precautions.
7. **Annexures** – site plan with BH locations, borelogs, test data sheets, graphs, photos.

Important Questions (Unit I)

1. What is the need for soil exploration? Explain the stages and planning of an exploration programme. (5–10 M)
2. State the depth & spacing of boreholes for (a) building, (b) highway, (c) dam.
3. Describe the different methods of boring with neat sketches – auger, wash, percussion, rotary.
4. Explain the **geophysical methods** – seismic refraction and electrical resistivity. Mention merits and demerits.
5. Differentiate disturbed and undisturbed samples. How are undisturbed samples obtained?
6. Define **area ratio, inside clearance, outside clearance**. What values are recommended for undisturbed sampling?
7. Describe the SPT with sketch. List the corrections applied to N and their formulas. (very frequent)
8. Write short notes on DCPT (type A and B) and static cone test; compare with SPT.
9. Explain the vane shear test and derive $c_u = T / [\pi (D^2H/2 + D^3/6)]$.
10. Describe PLT procedure and how load-settlement curve is used; give size-effect formulas. State limitations.
11. Explain the pressuremeter test and the parameters obtained.
12. List the contents of a soil investigation report. Draw a typical borelog.

Numerical practice

- SPT N = 40 at 8 m in fine saturated sand: find N_c, N' (γ = 20, GWT at GL, take γ_w = 10).

- Sampler: $D_o = 54$, $D_i (D_s) = 50$, $D_e = 49$ mm, $D_w = 53$ mm … find A_r, C_r, C_c.
- Vane $D = 50$ mm, $H = 100$ mm, $T = 14$ N·m → c_u and sensitivity (remoulded $T = 3.5$ N·m).
- PLT 0.45 m plate: 25 mm settlement at 200 kN/m² in sand; find settlement of 3 m footing at same pressure.

★ Quick revision: SPT = 63.5 kg × 750 mm, N = last 300 mm · Cone = 60°, 10 cm², 20 mm/s · Vane $c_u = T/3.66D^3$ · PLT depth of influence = $2B_p$ · $A_r \leq 10\ \%$

UNIT – II : EARTH SLOPE STABILITY

1. Introduction

- A **slope** = an inclined ground surface – natural (hill, valley side) or man-made (embankment, cutting, earth dam).
- Soil mass on a slope tends to slide down under its own weight (gravity). Slope stability analysis checks whether the shear strength available along a possible slip surface is enough to resist this.
- Aim of analysis: find the Factor of Safety (F) and the most dangerous (critical) slip surface (one with minimum F).

F = Shear strength available (τ_f) / Shear stress mobilised (τ_m)

- $F < 1 \rightarrow$ failure; $F = 1 \rightarrow$ limiting equilibrium; **F = 1.25 – 1.5** is generally adopted for permanent slopes (1.3 – 1.5 for earth dams; 1.0 – 1.2 temporary).

2. Classification of slopes

- **Infinite slope** – constant inclination β, very large extent, ends ignored; failure plane shallow & parallel to surface (long hill side).
- **Finite slope** – limited height H with crest & toe; slip surface curved/plane (embankments, dams, cuttings).

3. Causes of slope failure

(a) Increase in shear stress

- Extra load on crest; steepening of slope
- Removal of toe support (excavation, river erosion)
- Rain saturation; earthquake, blasting

(b) Decrease in shear strength

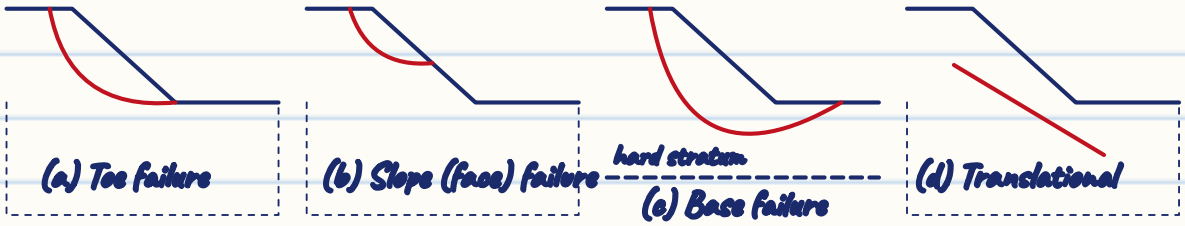
- Rise in pore water pressure u ($\sigma' = \sigma - u$ falls)
- Softening/swelling of clay, loss of c
- Weathering; progressive failure; sudden drawdown

Rule: any process that raises τ or lowers s $\Rightarrow$ F decreases $\Rightarrow$ slide!

4. Assumptions in stability analysis

- Problem treated as **plane strain** (2-D) – unit length along the slope.
- Homogeneous, isotropic soil; Mohr–Coulomb $s = c + \sigma' \tan\varphi$; soil above slip surface moves as a rigid body.

5. Types of failures of finite slopes



(a),(b),(c) = rotational slip (circular arc) – homogeneous cohesive soils
(d) = plane/wedge slip – along a weak layer, or c-φ=0 granular soil

Fig. 1 – Types of slope failure (red = slip surface)

(i) Rotational failure (circular arc)

- **Toe failure** – arc passes through the toe. Common in steep slopes ($\beta > 53°$) and soils with $\varphi > 0$.
- **Slope / face failure** – arc cuts the slope above toe; occurs when a hard layer exists at the toe level or for flat slopes with firm soil near the toe.
- **Base (deep) failure** – arc passes below the toe into the foundation; occurs in soft clays ($\varphi = 0$) with flatter slopes ($\beta < 53°$) with no hard stratum nearby.

(ii) Translational failure

- Slip surface is almost **plane**, parallel to the slope, at a shallow depth – infinite slopes, or when a weak stratum (soft clay seam, rock surface) is present.

(iii) Compound / wedge failure

- Curved + plane portions when a hard stratum lies at some depth; wedge sliding along 2–3 planes (rock slopes, weak seams).

Depth factor $n_d = D/H$; depth D of firm stratum below toe (H = height of slope). Base failure when D is large.

6. Methods of analysis

- Infinite slope: element equilibrium. Finite planar: Culmann. Finite circular: Swedish circle ($\varphi=0$), slices (Fellenius/Bishop), friction circle, Taylor charts.

INFINITE SLOPES

7. Infinite slope in cohesionless soil (dry)

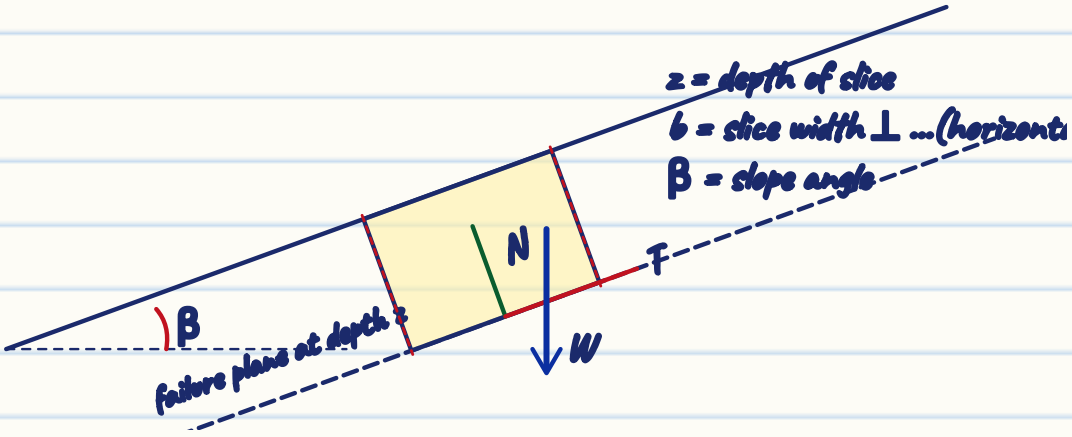


Fig. 2 – Forces on a slice of an infinite slope

- Consider a slice of width b (horizontal) and depth z. Side forces on the two vertical faces are equal & opposite (same depth, same conditions) – they cancel.
- Weight of slice: $W = \gamma\, b\, z$ (per unit length).
- Length of failure plane: $l = b / \cos\beta$.

$$N = W\cos\beta = \gamma\, b\, z\cos\beta \;;\; T = W\sin\beta = \gamma\, b\, z\sin\beta$$

Normal stress $\sigma = N/l = \gamma\, z\cos^2\beta$; Shear stress $\tau = T/l = \gamma\, z\sin\beta\cos\beta$

For cohesionless dry soil: $s = \sigma\tan\varphi = \gamma\, z\cos^2\beta\tan\varphi$

$$F = s/\tau = (\gamma\, z\cos^2\beta\tan\varphi)/(\gamma\, z\sin\beta\cos\beta) = \tan\varphi / \tan\beta$$

- F is **independent of depth** z and of unit weight γ.
- Limiting stable slope: $F = 1 \Rightarrow \beta = \varphi$ (angle of repose). A dry sand slope can never be steeper than φ.
- For $F = 1.5$ and $\varphi = 30°$ $\Rightarrow \tan\beta = 0.5774/1.5 \Rightarrow \beta = 21°$.

Same answer if the slope is completely submerged in still water (no seepage) – because both γ' and u act, ratio unchanged.

8. Infinite slope in c–φ soil (no seepage)

- Shear strength: $s = c + \sigma\tan\varphi = c + \gamma\, z\cos^2\beta\tan\varphi$
- Mobilised shear stress: $\tau = \gamma\, z\sin\beta\cos\beta$

$$F = c/(\gamma\, z\sin\beta\cos\beta) + \tan\varphi/\tan\beta$$

$$F = F_c + F_\varphi \;,\; F_c = c/(\gamma\, z\cos^2\beta\tan\beta)$$

- F is made of a **cohesion part** (decreases with depth z) and a **friction part** (independent of z).
- F reduces as z increases ⇒ deeper failure planes are more critical for c–φ soil until a rock/firm layer.

Critical depth (Hc) – depth at which F = 1

$$1 = c/(\gamma H_c \cos^2\beta (\tan\beta)) + \tan\varphi/\tan\beta$$

$$H_c = c / [\gamma \cos^2\beta (\tan\beta - \tan\varphi)]$$

Stability number (S_n) for infinite slope

$$S_n = c / (\gamma H_c) = \cos^2\beta (\tan\beta - \tan\varphi)$$

- For cohesive soil φ = 0: $H_c = c/(\gamma \sin\beta \cos\beta) = 2c / (\gamma \sin 2\beta)$.
- Hence for given F, the safe depth: $H = c / [\gamma \cos^2\beta (\tan\beta - \tan\varphi_m)]$ where $c_m = c/F_c$, $\tan\varphi_m = \tan\varphi/F_\varphi$ are mobilised values ($F_c = F_\varphi = F$).

9. Infinite slope with seepage parallel to the slope

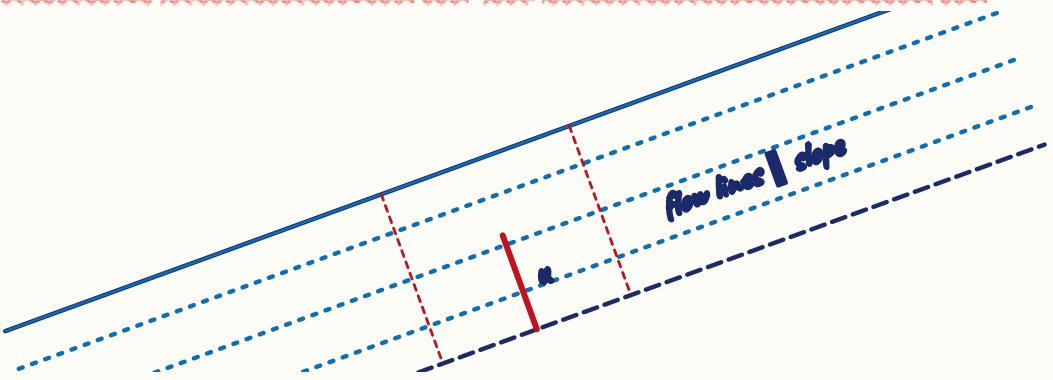


Fig. 3 – Water table at ground surface, flow parallel to slope

- Flow lines ∥ slope, equipotentials ⊥ slope. Pore pressure on the failure plane = water head along equipotential: $u = \gamma_w z \cos^2\beta$.
- Saturated unit weight γ_{sat} used for W: $\sigma = \gamma_{sat} z \cos^2\beta$; $\sigma' = \sigma - u = (\gamma_{sat} - \gamma_w) z \cos^2\beta = \gamma' z \cos^2\beta$.

$$F = c' / (\gamma_{sat} z \sin\beta \cos\beta) + (\gamma'/\gamma_{sat}) \cdot (\tan\varphi'/\tan\beta)$$

- Since $\gamma' \approx \gamma_{sat}/2$, the friction term is nearly halved – seepage parallel to slope reduces safety by about 50 % for cohesionless soil.
- For c' = 0: $F = (\gamma'/\gamma_{sat}) \tan\varphi'/\tan\beta$ ⇒ stable slope angle is only about **half of φ'**: $\tan\beta \approx ½ \tan\varphi'$.

- If the water table is at depth m z (m = fraction of z that is saturated): $u = \gamma_w (1-m)$... $F = c'/(\gamma z \sin\beta \cos\beta) + [1 - r_u \sec^2\beta] \tan\varphi'/\tan\beta$ where $r_u = u/\gamma z$ (pore pressure ratio).

Condition (c = 0)	F
Dry slope	$\tan\varphi/\tan\beta$
Fully submerged, no seepage	$\tan\varphi/\tan\beta$
Seepage ∥ slope, WT at surface	$(\gamma'/\gamma_{sat}) \tan\varphi/\tan\beta \approx \frac{1}{2} \tan\varphi/\tan\beta$

Important: dry/submerged → $\beta \leq \varphi$. With seepage → $\beta \leq \sim \varphi/2$! This is why slopes fail after heavy rain.

Example 1

An infinite slope of inclination 20° is made of soil with $c' = 10$ kN/m², $\varphi' = 30°$, $\gamma_{sat} = 19$ kN/m³. Find F against a failure plane at depth 3 m parallel to slope if there is seepage parallel to slope with water table at ground surface. ($\gamma_w = 9.81$)

Solution: $\gamma' = 19 - 9.81 = 9.19$ kN/m³; $\sin20° = 0.342$, $\cos20° = 0.940$, $\tan20° = 0.364$, $\tan30° = 0.577$

$$F = 10/(19\times3\times0.940\times0.342) + (9.19/19)(0.577/0.364)$$
$$= 0.546 + 0.484 \times 1.586 = 0.546 + 0.767$$

F = 1.31 (safe, > 1.25 ✓)

Check without seepage: $F = 0.546 + 1.586 \times (19/19) = 2.13$ (using $\gamma_{sat} = \gamma$) – the seepage almost halves F (2.13 → 1.31).

Exam tip: write the formula, substitute, and conclude "slope is safe / unsafe" as per $F \geq 1.25$.

FINITE SLOPES – CIRCULAR SLIP

10. Swedish Circle Method ($\varphi_u = 0$ analysis)

- Developed by Swedish geotechnical commission (Petterson, Fellenius) after the Gota-river quay failures (1916).
- Suitable for **saturated clay under undrained (short-term) condition:** $\varphi_u = 0$, shear strength = c_u (constant).

- Slip surface = circular arc; whole soil mass above it taken as one block rotating about centre O.

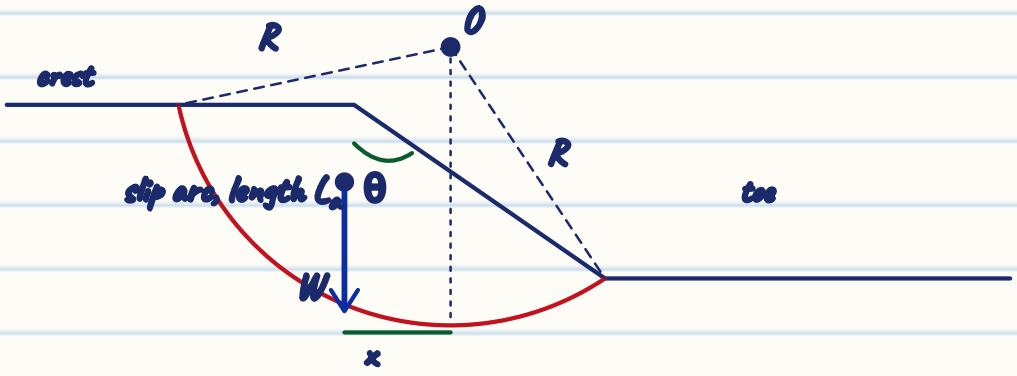


Fig. 4 – Swedish circle ($\varphi = 0$)

Derivation

- W = weight of the sliding mass (area × γ, per unit length), acts at its centroid, lever arm x from O → **Disturbing moment** $M_D = W \cdot x$
- Resisting force on arc = $c_u \times L_a$ where L_a = arc length = $R\theta$ (θ in radians) acts at distance R → **Resisting moment** $M_R = c_u L_a R = c_u R^2\theta$

$$F = M_R/M_D = c_u L_a R / (W x) = c_u R^2\theta / (W x)$$

- Trial circles are repeated with different centres; the smallest F = critical circle (F_{min}).

Tension cracks

- In cohesive soil a vertical **tension crack** forms at the crest to depth $z_c = 2c_u/\gamma$ (tensile strength = 0).
- Effect: (i) shear resistance along crack length is lost → arc length reduces to L_a' (arc only up to crack) and W reduces; (ii) if crack fills with water, a hydrostatic thrust $P_w = \frac{1}{2}\gamma_w z_c^2$ acts horizontally at $z_c/3$ from the crack bottom, adding to disturbing moment.

$$F = c_u L_a' R / (W x + P_w y_w)$$

- y_w = vertical distance from O to the line of action of P_w. Cracks reduce F → hence the critical circle must be tried with cracks.

Limitations of Swedish circle method

- Applicable only for $\varphi = 0$ soils (undrained clays).

- Does not consider that c_u may vary with depth, so a layered soil must be dealt with by slices.
- Trial-and-error: many circles required.

For c–φ soil, use method of slices (next pages).

Example 2 (Swedish circle)

A trial slip circle of radius 10 m is selected for an embankment on saturated clay (c_u = 30 kN/m², φ_u = 0). The area of the sliding mass is 35 m², γ = 20 kN/m³; centroid of the sliding mass is at horizontal distance 4.5 m from the centre O. The central angle of the arc is 103°. Find F.

Solution

- $W = A \times \gamma = 35 \times 20 =$ **700 kN/m**
- $\theta = 103° = 103 \times \pi/180 = 1.798$ rad ; $L_a = R\theta = 10 \times 1.798 =$ **17.98 m** ≈ 18 m
- $M_R = c_u L_a R = 30 \times 17.98 \times 10 = 5394$ kN·m/m
- $M_D = W x = 700 \times 4.5 = 3150$ kN·m/m

$$F = 5394 / 3150 = 1.71 > 1.5 \text{ (safe)}$$

If a water-filled tension crack of $z_c = 2\times30/20 = 3$ m existed: $P_w = ½ \times 9.81 \times 3^2 = 44$ kN/m (arm ≈ 8 m) → M_D increases by 352 → $F \approx 1.54$ (reduced).

II. Method of slices – Standard (Fellenius / Swedish) method

- For c–φ soils, non-homogeneous slopes, with seepage. Sliding mass above arc divided into **n vertical slices** of equal width b (≈ R/10).
- Each slice treated separately: forces on it –

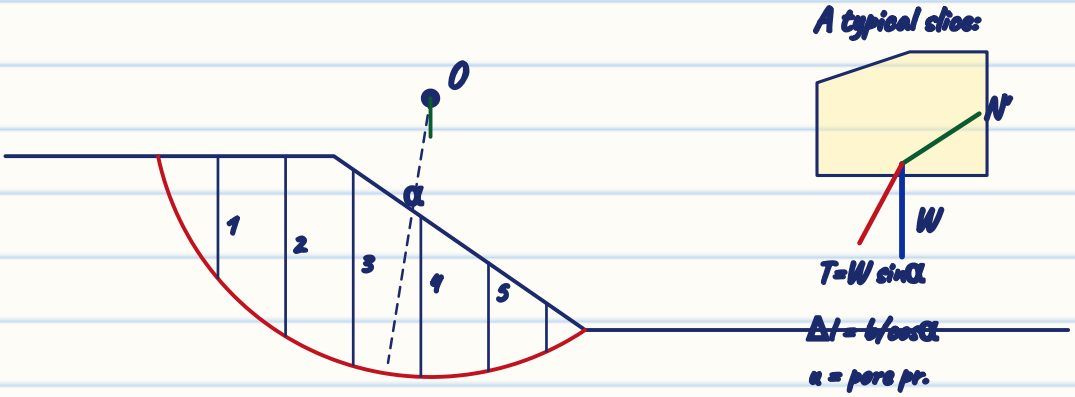


Fig. 5 – Slices (b = width, α = angle of base with horizontal)

- W (weight) ; $N = W\cos\alpha$ (normal) ; $T = W\sin\alpha$ (tangential). Inter-slice forces neglected (Fellenius assumption: resultant of side forces is parallel to the slice base) – so no change to the above.
- Slice base length $\Delta l = b / \cos\alpha$. α is +ve when base slopes up in same direction as the slope surface (toward toe), –ve on the other side of the vertical through O.

Derivation (effective stress)

- Shear resistance on slice base: $S = [\, c'\Delta l + (N - u\Delta l)\tan\varphi' \,] / F$
- Moments about O: driving = $\Sigma\, W R \sin\alpha$; resisting = $R \Sigma\, S$"

$$F = \Sigma\, [\, c'\Delta l + (W\cos\alpha - u\Delta l)\tan\varphi' \,] / \Sigma\, W\sin\alpha$$

- Total stress / undrained case ($\varphi = 0$, u not considered): $F = \Sigma\, c_u \Delta l / \Sigma\, W\sin\alpha = c_u L_a / \Sigma\, W\sin\alpha$. (Equivalent to Swedish circle)
- Pore pressure ratio: $r_u = u/(\gamma h) = u\, b / W \Rightarrow u\Delta l = r_u W / \cos\alpha$ … so $F = \Sigma[c'\Delta l + W(\cos\alpha - r_u \sec\alpha)\tan\varphi\,]/\Sigma\, W\sin\alpha$

Fellenius: simple, conservative (F is 5–20 % lower than Bishop), but can be too conservative for large pore pressures/flat arcs.

12. Bishop's simplified method (mention)

- Considers inter-slice normal forces E; assumes vertical shear forces between slices are zero ($X_n - X_{n+1} = 0$). Vertical equilibrium of each slice gives N' so that F appears on both sides – **iterative**.

$$F = \Sigma\, \{\, [\, c'b + (W - ub)\tan\varphi\,] / m_\alpha \,\} / \Sigma\, W\sin\alpha$$

$$m_\alpha = \cos\alpha\,(1 + \tan\alpha\tan\varphi'/F)$$

- Assume F (take Fellenius value) → calculate m_α → new F → repeat until same (2–3 trials).

Feature	Fellenius (ordinary)	Bishop simplified
Inter-slice forces	Ignored	Normal force E included, shear X = 0
Equations satisfied	Moment only	Moment + vertical force
Solution / result	Direct, conservative	Iterative, more accurate

Steps for method of slices

1. Draw the slope to scale; select a trial centre O and radius R; draw circle.
2. Divide into 6–10 slices (equal widths; a slice must have a single base soil).
3. Find $W = \gamma \times$ area, α, Δl, u (flow net) for each; tabulate $W\cos\alpha$, $W\sin\alpha$, $c'\Delta l$.
4. Compute F; repeat for other circles; plot contours of F – minimum F = critical.

Example 3 (Fellenius method)

For a trial circle, the sliding mass is divided into 5 slices of width 2 m. $c' = 10\ kN/m^2$, $\varphi' = 20°$, no pore pressure ($u = 0$). Find F.

Slice	W (kN/m)	α	$\Delta l = 2/\cos\alpha$	$W\cos\alpha$	$W\sin\alpha$
1	40	−8°	2.02	39.6	−5.6
2	90	6°	2.01	89.5	9.4
3	120	20°	2.13	112.8	41.0
4	95	35°	2.44	77.8	54.5
5	40	50°	3.11	25.7	30.6
Σ			**11.71**	**345.4**	**130.0**

$c\Sigma\Delta l = 10 \times 11.71 = 117.1$; $\Sigma W\cos\alpha\tan\varphi' = 345.4 \times 0.364 = 125.7$

$$F = (117.1 + 125.7)/130.0 = 1.87$$

FRICTION CIRCLE METHOD

13. Friction circle method (Taylor)

- A **graphical**/force-polygon method for c–φ soil (homogeneous), treats the slip mass as a whole (no slices). Developed by Taylor (1937).
- Three forces act on the sliding mass at failure: (1) **W** – weight through centroid, (2) **C** – resultant of mobilised cohesion on the arc, (3) **P** – resultant of normal & friction forces on arc.

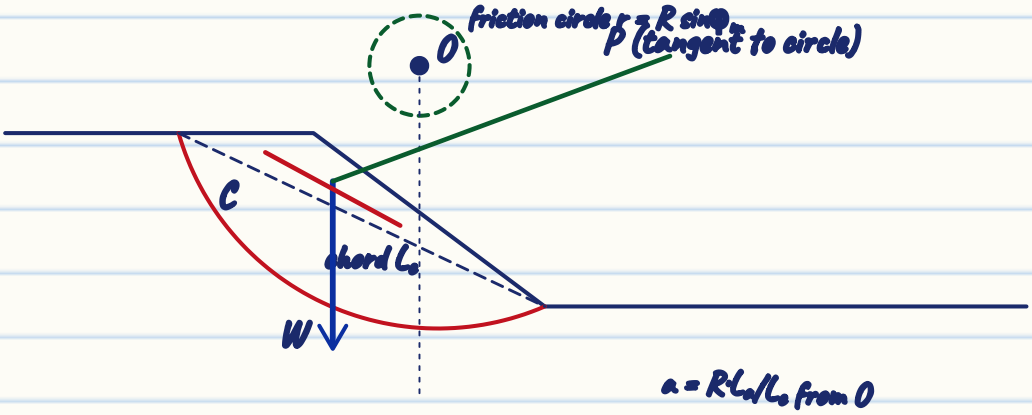


Fig. 6 – Friction circle method

Concept

- Mobilised friction angle $\varphi_m = \tan^{-1}(\tan\varphi / F_\varphi)$; mobilised cohesion $c_m = c / F_c$.
- The resultant P of normal & frictional forces at any point of the arc makes angle φ_m with the normal ⇒ it is tangent to a circle of radius $r_\varphi = R \sin\varphi_m$ concentric with slip circle – the **friction circle**. (Line of P assumed tangent to this circle.)
- Resultant cohesion force: $C = c_m \times L_c$ (L_c = chord length), parallel to chord, at perpendicular distance from O:

$$a = R \cdot L_a / L_c \ (L_a = \text{arc length})$$

Procedure (graphical)

1. Draw slope to scale. Choose trial circle (centre O, radius R); find area & centroid ⇒ $W = \gamma A$.
2. Assume a trial factor of safety against friction F_φ (say 1.5); calculate $\varphi_m = \tan^{-1}(\tan\varphi / F_\varphi)$ and draw friction circle of radius $R \sin\varphi_m$.
3. Draw chord AB, measure L_c; calculate arc length L_a; find $a = R\, L_a / L_c$. Draw line of action of cohesion C parallel to AB at distance a from O.
4. Extend W and C to intersect at a point; draw the line of P through this point tangent to friction circle (so that the moment of P about O has the same sense as W's, i.e. P resists).
5. Draw force triangle: W known in magnitude and direction; draw C ∥ chord and P along tangent line → closed triangle gives the value of C_m (mobilised cohesion force).
6. $c_m = C_m / L_c$; $F_c = c / c_m$.

7. If $F_c \neq F_\varphi$ assumed, repeat with new F_φ until $F_c = F_\varphi = F$ (plot F_c vs F_φ and read intersection with 45° line).

Special case $\varphi = 0$: friction circle vanishes ($r = 0$) → P passes through O → reduces to Swedish circle: $F = c\, L_a\, R/(W\, x)$.

Merits & limitations

Merits

- Whole mass treated – no slices
- Handles c–φ soil, with seepage by using effective stress & total weight
- Basis for Taylor's charts

Limitations

- Assumes homogeneous soil
- Trial & error and tedious graphics
- Tangent assumption of P is approximate (slightly conservative)

14. Taylor's stability number method

- Taylor (1948) analysed homogeneous slopes by friction-circle method and presented results as **dimensionless charts**, so no tedious trial circles are needed.
- Define the stability number :

$$S_n = c_m / (\gamma H) = c / (F_c \gamma H)$$

- c_m = mobilised (required) cohesion; H = height of slope; γ = unit weight. S_n depends on slope angle β, mobilised friction angle φ_m and depth factor $n_d = D/H$.
- Reciprocal $1/S_n = \gamma H / c_m$ is called **stability factor** (m). Critical height ($F = 1$): $H_c = c/(\gamma S_n)$.

Taylor's chart for $\varphi = 0$ (approx. values of S_n)

Slope β	90°	75°	60°	53°	45°	30°	15°
S_n	0.261	0.219	0.191	0.181	0.170	0.156	0.145

- $\beta > 53°$: always toe failure; S_n independent of depth factor.
- $\beta < 53°$: failure may be toe, slope or base type depending on depth factor $n_d = D/H$ (chart gives curves for n_d = 1, 1.25, 1.5, 2 … ∞). Maximum S_n for $\varphi = 0$ is 0.181 at $\beta = 53°$ ($n_d = \infty$ → midpoint circle).
- For $\varphi > 0$ (φ = 5°, 10°, 15°, 20°, 25°) curves of S_n vs β are given (toe circles) – S_n decreases as φ increases.

Procedure for c–φ soil (stability analysis)

1. Assume F_φ ; $\varphi_m = \tan^{-1}(\tan\varphi / F_\varphi)$.

2. From chart with β, φ_m (and n_d) read S_n.

3. $c_m = S_n \gamma H \Rightarrow F_c = c/c_m$.

4. Repeat until $F_c = F_\varphi = F$.

Design problem: given F, find H or β: $c_m = c/F$; $\varphi_m = \tan^{-1}(\tan\varphi/F) \Rightarrow S_n$ from chart $\Rightarrow H = c_m/(\gamma S_n) = c/(F\gamma S_n)$.

If only F against cohesion is wanted ($\varphi = 0$): $F = c / (S_n \gamma H)$ – straight, no trial!

Example 4 (Taylor)

A 10 m high cut slope of 1 H : 1 V ($\beta = 45°$) is in saturated clay with $c_u = 40$ kN/m², $\varphi_u = 0$, $\gamma = 19$ kN/m³. A hard stratum is at great depth ($n_d = \infty$). Find F and critical height.

Solution: Taylor's chart $\varphi = 0$, $\beta = 45° \Rightarrow S_n = 0.170$.

$$c_m = S_n \gamma H = 0.170 \times 19 \times 10 = 32.3 \text{ kN/m}^2$$

$$F = c/c_m = 40/32.3 = 1.24$$

$$H_c = c/(\gamma S_n) = 40/(19 \times 0.170) = 12.4 \text{ m}$$

EARTH DAMS – STABILITY OF SLOPES

15. Stability of slopes of earth dams

- Earth dam = embankment of compacted soil to store water; slopes must be stable **at all stages of life:** construction, reservoir full, drawdown.
- Shear strength parameters and pore pressures differ at each stage ⇒ four critical conditions are checked, using method of slices (circular arcs) – trial circles till F_{min}.

Condition	Slope	Analysis	Min F
(a) End of construction	Upstream & downstream	Total stress (UU) or effective with construction u	1.25
(b) Steady seepage (reservoir full)	**Downstream**	Effective stress (c', φ'), u from flow net	1.5
(c) Sudden drawdown	**Upstream**	Effective stress, u of full pool retained	1.2–1.3
(d) Partial pool	Upstream	Effective stress	1.5

(b) Steady seepage condition

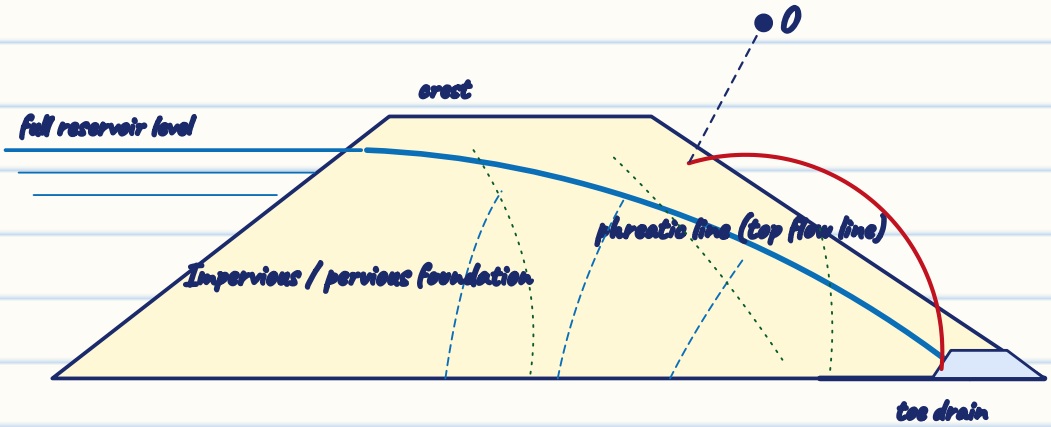


Fig. 7 – Earth dam, steady seepage : phreatic line, flow net & critical slip circle on d/s side

- Phreatic line (top flow line) = **Casagrande's parabola** corrected at entry; flow net drawn; pore pressure at base mid-point of each slice: $u = \gamma_w h_w$ (h_w = height from point up to the equipotential line meeting the point, measured to the phreatic level).
- Soil **above phreatic line:** moist γ ; **below:** saturated γ_{sat} (use in W). Shear strength in terms of **effective stress** (c', φ') from CD/CU tests:

$$F = \Sigma \left[c'\Delta l + (W\cos\alpha - u\Delta l)\tan\varphi' \right] / \Sigma\, W\sin\alpha$$

- Downstream slope is the critical one – seepage force acts outward; pore pressures are high near the toe ⇒ provide **toe drain / rock toe** and a filter to lower the phreatic line ⇒ F increases (typically from 1.1 to > 1.5).
- Alternatively use r_u (average pore pressure ratio = 0.3–0.5): $u\Delta l = r_u W \sec\alpha$.

(a) End of construction condition

- Pore pressures generated by compaction/loading have not yet dissipated (especially in clayey cores) – **undrained.**
- Analyse in **total stress** with UU test parameters (c_u, φ_u) – weights with moist γ: $F = \Sigma[c_u\Delta l + W\cos\alpha\tan\varphi_u]/\Sigma\, W\sin\alpha$. For saturated clay $\varphi_u = 0$ (Swedish circle).
- Or use effective-stress parameters with u from construction pore pressure (r_u from piezometers). Both upstream and downstream slopes checked; critical for dams on soft foundation clays.
- Reservoir empty ⇒ no water force on u/s face; stability improves later with consolidation (F ↑ with time).

(c) Sudden (rapid) drawdown condition

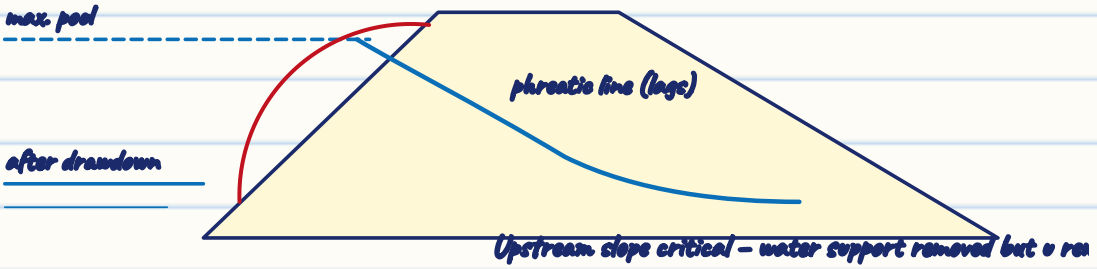


Fig. 8 – Sudden drawdown

- Reservoir level falls quickly (faster than soil can drain – in low-permeability shell/core). The water pressure on the u/s face disappears but pore pressures inside remain at **full-reservoir level** (no dissipation).
- Soil saturated ⇒ W uses γ_{sat} for all submerged soil (no buoyancy). Normal stress ($\sigma - u$) reduced and driving force increased ⇒ most critical for the upstream slope.
- Pore pressure can be obtained from the steady seepage flow net (before drawdown) – effective-stress method: $F = \Sigma[c'\Delta l + (W\cos\alpha - u\Delta l)\tan\varphi']/\Sigma W\sin\alpha$ with W from γ_{sat}; OR use the CU (consolidated-undrained) parameters in total stress analysis: $F = \Sigma[c_{cu}\Delta l + W\cos\alpha\tan\varphi_{cu}]/\Sigma W\sin\alpha$ (Bishop's / U.S. Army Corps two-stage method).
- Morgenstern (1963) charts give F directly for drawdown ratio ($\Delta H/H$); F is lowest when drawdown is quick and the soil is of low permeability (clay).
- Free-draining sand / gravel shells dissipate u quickly ⇒ drawdown is not critical for these.

Remedies: flatter u/s slope, rip-rap, pervious shell, controlled drawdown rate (≈ 15 cm/day).

Summary of key formulae

Case	Factor of safety
Infinite slope, $c = 0$	$\tan\varphi / \tan\beta$
Infinite slope, c–φ	$c/(\gamma z \sin\beta \cos\beta) + \tan\varphi/\tan\beta$
Infinite, seepage ∥ slope	$c'/(\gamma_{sat} z \sin\beta \cos\beta) + (\gamma'/\gamma_{sat}) \tan\varphi'/\tan\beta$
Swedish circle ($\varphi=0$)	$c_u L_a R/(W x)$
Fellenius slices	$\Sigma[c'\Delta l+(W\cos\alpha-u\Delta l)\tan\varphi']/\Sigma W \sin\alpha$
Bishop	$\Sigma\{[c'b+(W-ub)\tan\varphi']/m_\alpha\}/\Sigma W \sin\alpha$
Taylor	$F_c = c/(S_n \gamma H)$; $H_c = c/(\gamma S_n)$

Important questions (exam)

1. Explain the different types of slope failures with neat sketches. What are the causes of failure of slopes?
2. Derive the expression for factor of safety of an infinite slope in c–φ soil. How is it modified for seepage parallel to the slope?
3. Explain Swedish circle method for analysing stability of a slope in purely cohesive soil.
4. Describe the method of slices (Fellenius) for slope stability. Compare with Bishop's method.
5. Explain the friction circle method with neat sketch. Define Taylor's stability number and explain its use.
6. Discuss the stability of u/s and d/s slopes of an earth dam under (i) steady seepage (ii) sudden drawdown (iii) end of construction.
7. Numericals: infinite slope with seepage; Swedish circle with tension crack; Fellenius table; Taylor's H_c and F.

UNIT – III : EARTH PRESSURE THEORIES & RETAINING WALLS

1. Introduction – Lateral Earth Pressure

- Retaining structures (walls, sheet piles, abutments, basement walls, bunkers) hold back soil → soil pushes **sideways** on them. This horizontal thrust = lateral earth pressure.
- Magnitude depends on: type of soil (c, φ, γ), wall movement (yielding / rigid), wall friction (δ), backfill slope, surcharge, water table, drainage.
- Pressure intensity at depth z : $\sigma_h = K \cdot \sigma_v$, K = coefficient of earth pressure.

2. Three States of Earth Pressure

State	Wall movement	Coefficient
At-rest	None (rigid, unyielding wall) – zero lateral strain	K_0
Active	Wall moves away from soil → soil expands, shear strength fully mobilised	K_a (minimum)
Passive	Wall pushed towards soil → soil compressed	K_p (maximum)

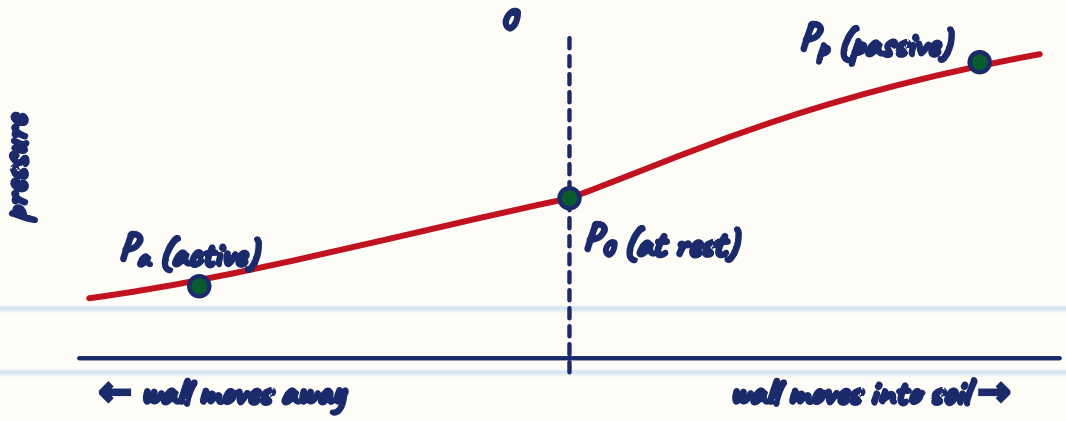


Fig 1 : Lateral pressure vs wall movement

- Movement needed to mobilise: **active** ≈ 0.001 H (dense sand) – 0.004 H (loose sand), **passive** ≈ 0.01 H – 0.05 H (passive needs ≈ 10 times more movement).
- Order : $K_a < K_0 < K_p$

3. Coefficient of Earth Pressure At Rest (K_0)

$$K_0 = \sigma_h/\sigma_v = \nu/(1-\nu) \quad \text{(elastic)}$$
$$K_0 = 1 - \sin\varphi' \text{ (Jaky, N.C. soil)}; \; K_{0(OC)} = K_{0(NC)}\sqrt{OCR}$$

- Typical K_0: loose sand 0.5, dense sand 0.4, N.C. clay 0.5–0.65, over-consolidated clay up to 1.0 or more.
- Pressure at rest: $p_0 = K_0 \gamma z$; thrust $P_0 = \tfrac{1}{2} K_0 \gamma H^2$ acting at $H/3$ from base.

4. Rankine's Theory of Earth Pressure (1857)

Assumptions

- Soil is **homogeneous, isotropic, semi-infinite** and cohesionless (extended later to c-φ soil).
- Wall is **vertical, smooth** (no wall friction, $\delta = 0$) and rigid; the back of the wall does not disturb the stress field.
- Ground surface is plane; plastic equilibrium exists throughout the backfill; for level ground the thrust is **horizontal**.

Active case – cohesionless soil, horizontal backfill

- Vertical stress $\sigma_v = \gamma z$ = **major** principal stress σ_1; horizontal stress $\sigma_h = \sigma_3$ (minor).
- At failure, Mohr circle touches the envelope $\tau = \sigma \tan\varphi$:

$$\sin\varphi = (\sigma_1 - \sigma_3)/(\sigma_1 + \sigma_3) \Longrightarrow \sigma_3 = \sigma_1 (1 - \sin\varphi)/(1 + \sin\varphi)$$

$$\sigma_a = K_a \gamma z, \quad K_a = (1 - \sin\varphi)/(1 + \sin\varphi) = \tan^2(45^\circ - \varphi/2)$$

$$P_a = \tfrac{1}{2} K_a \gamma H^2 \text{ acting at } H/3 \text{ above base}$$

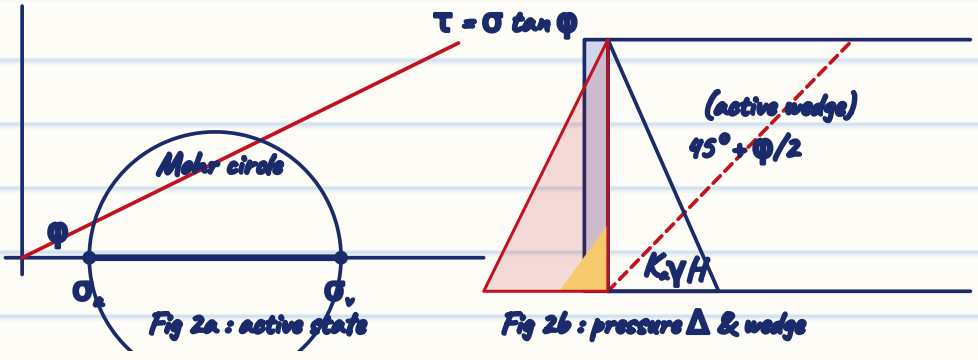


Fig 2a : active state Fig 2b : pressure Δ & wedge

- Failure (slip) planes make $(45° + \varphi/2)$ with the horizontal in the active state; two sets of planes inclined at $\pm (45° + \varphi/2)$.

Passive case – cohesionless soil, horizontal backfill

- Now σ_h becomes the **major** principal stress, $\sigma_v = \gamma z$ the minor.

$$\sigma_p = K_p \gamma z, \quad K_p = (1 + \sin\varphi)/(1 - \sin\varphi) = \tan^2(45° + \varphi/2) = 1/K_a$$

$$P_p = \tfrac{1}{2} K_p \gamma H^2 \text{ at } H/3 \text{; slip planes at } (45° - \varphi/2) \text{ to horizontal}$$

For $\varphi = 30°$: $K_a = 1/3$, $K_p = 3$, $K_0 = 0.5$

5. Rankine's Theory – Cohesive (c–φ) Soil

- Mohr–Coulomb : $\tau = c + \sigma \tan\varphi$. From the Mohr circle:

$$\sigma_a = \gamma z K_a - 2c\sqrt{K_a} \quad / \quad \sigma_p = \gamma z K_p + 2c\sqrt{K_p}$$

- At the surface ($z = 0$) active pressure $= -2c\sqrt{K_a}$ (tension) → soil tends to separate from the wall → **tension cracks.**
- Depth of tension crack ($\sigma_a = 0$):

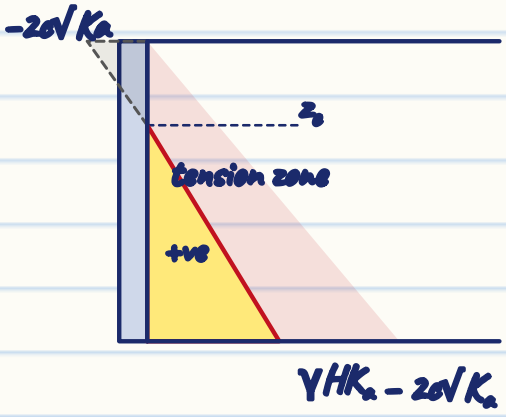
$$z_c = 2c / (\gamma \sqrt{K_a}) = (2c/\gamma) \tan(45° + \varphi/2)$$

- Depth up to which unsupported vertical cut stands (σ_a net = 0 over H): $H_c = 2 z_c = 4c/(\gamma\sqrt{K_a})$.
- For $\varphi = 0$ clay : $K_a = 1 \Longrightarrow \sigma_a = \gamma z - 2c$; $z_c = 2c/\gamma$.

Total active thrust (c–φ soil, height H)

$$\text{Without crack : } P_a = \tfrac{1}{2}\gamma H^2 K_a - 2cH\sqrt{K_a} + 2c^2/\gamma$$

$$\text{With crack : } P_a = \tfrac{1}{2}\gamma K_a (H - z_c)^2 \text{, acts at } (H - z_c)/3 \text{ above base}$$



- Net pressure negative upto z_c
- Pressure diagram: –ve triangle + +ve triangle
- Tension is ignored (soil cannot pull wall) → only +ve part used

Fig 3 : Active pressure in c–φ soil

6. Effect of Uniform Surcharge (q per unit area)

- Surcharge q adds to vertical stress at every depth : $\sigma_v = q + \gamma z$.
- Active pressure : $\sigma_a = K_a(q + \gamma z)$ → trapezoidal diagram. Equivalent height of extra fill $h_e = q/\gamma$.

$$P_a = K_a q H + \tfrac{1}{2} K_a \gamma H^2$$

- Line of action (from base) : $\bar{y} = [K_a q H \cdot (H/2) + \tfrac{1}{2} K_a \gamma H^2 \cdot (H/3)] / P_a$.
- For c–φ soil add $- 2c\sqrt{K_a}$ (constant) to every ordinate.

7. Effect of Water Table (backfill partly / fully submerged)

- Below water table use submerged unit weight $\gamma' = \gamma_{sat} - \gamma_w$ for soil pressure **plus** full hydrostatic pressure $\gamma_w z_w$ (water pressure is not reduced by K).

$$\sigma_{total} = K_a[\gamma H_1 + \gamma'(z - H_1)] + \gamma_w (z - H_1)$$

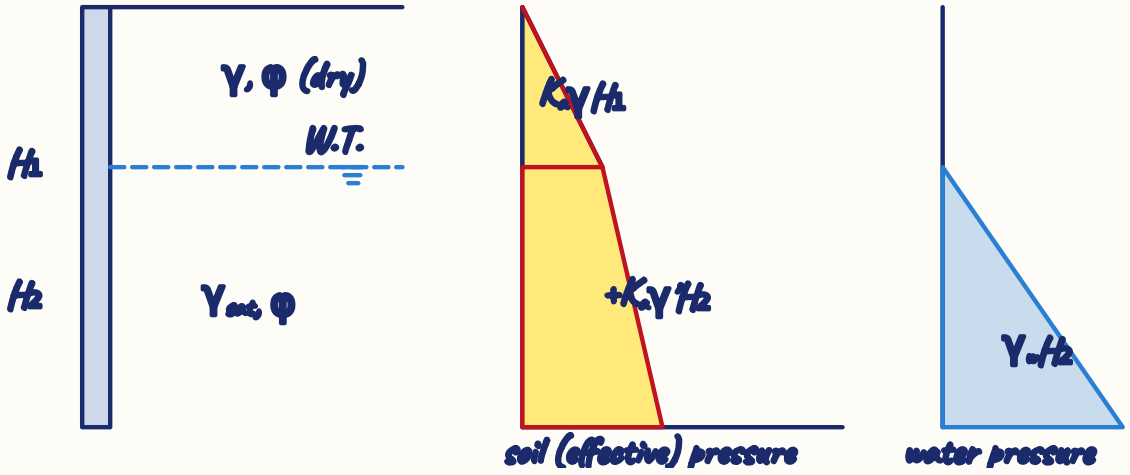


Fig 4 : Pressure diagram with water table at depth H_1

- Thrust = areas of triangle + rectangle + triangle (soil) + $\tfrac{1}{2}\gamma_w H_2^2$ (water). Water increases thrust nearly 2 times → *provide drainage!*
- Passive side with submerged soil: use $K_p \gamma' z + \gamma_w z$ similarly.

8. Inclined Backfill (angle β) – Rankine

$$K_a = \cos\beta \cdot [\cos\beta - \sqrt{(\cos^2\beta - \cos^2\varphi)}] / [\cos\beta + \sqrt{(\cos^2\beta - \cos^2\varphi)}]$$

- $P_a = \tfrac{1}{2} K_a \gamma H^2$ acts **parallel to the sloping surface** (inclined β to horizontal) at H/3. For β = 0 it reduces to $(1 - \sin\varphi)/(1 + \sin\varphi)$; need $\beta \le \varphi$.

9. Earth Pressure in Layered Soils (Rankine)

- Each layer has its own γ, c, φ → its own K_a. Pressure is **continuous in vertical stress σ_v** but jumps at the interface because K changes.
- **Procedure :**
  1. Compute σ_v at each layer boundary (bottom of layer 1 = $\gamma_1 H_1 + q$).
  2. Upper layer: $\sigma_a = K_{a1}\,\sigma_v$ (use above the interface) ; lower layer: $\sigma_a = K_{a2}\,\sigma_v$ (just below the interface).
  3. Below the water table use γ' and add $\gamma_w z_w$. For c-soil subtract $2c\sqrt{K_a}$ in that layer only.
  4. Split diagram into rectangles + triangles; P = areas ; $\bar{y} = \Sigma P{\cdot}y \,/\, \Sigma P$ (moment about base).

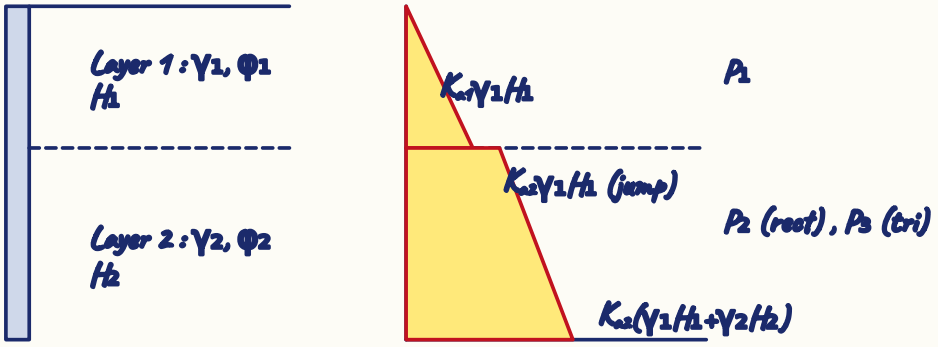


Fig 5 : Two-layer cohesionless backfill

If lower layer is clay ($\varphi = 0$): $K_{a2} = 1 \rightarrow \sigma_a = \sigma_v - 2c$; upper layer acts as surcharge $q = \gamma_1 H_1$.

Example 1 – Layered backfill

Q. A 6 m high smooth vertical wall retains horizontal backfill: 0–3 m γ = 18 kN/m³, φ = 30°; 3–6 m γ = 20 kN/m³, φ = 20°. Find total active thrust and its point of application.

$K_{a1} = (1 - \sin30°)/(1 + \sin30°) = 0.333$; $K_{a2} = (1 - \sin20°)/(1 + \sin20°) = 0.490$

Level	σ_v (kPa)	$\sigma_a = K\,\sigma_v$ (kPa)
0 m	0	0
3 m (above)	18×3 = 54	0.333×54 = 18.0
3 m (below)	54	0.490×54 = 26.5
6 m	54 + 20×3 = 114	0.490×114 = 55.9

Solution (contd.) – Thrusts per metre run & moments about base:

Part	Force (kN/m)	Arm (m)	Moment (kNm)
$P_1 = \frac{1}{2}\times 18\times 3$	27.0	3 + 1 = 4.0	108.0
$P_2 = 26.46\times 3$	79.4	1.5	119.1
$P_3 = \frac{1}{2}\times(55.86-26.46)\times 3$	44.1	1.0	44.1
Total	150.5		271.2

$P_a = 150.5$ kN/m ; $\bar{y} = 271.2/150.5 = 1.80$ m above base

Example 2 – c–φ soil with tension crack

Q. A vertical smooth wall 5 m high retains a clay backfill: $\gamma = 18$ kN/m^3, $c = 10$ kN/m^2, $\varphi = 20°$. Find (a) depth of tension crack (b) active thrust with & without crack, (c) its point of application.

$K_a = \tan^2 35° = 0.490$, $\sqrt{K_a} = 0.700$

- σ_a at top $= -2c\sqrt{K_a} = -2\times 10\times 0.7 =$ **– 14 kPa**
- σ_a at base $= 18\times 5\times 0.49 - 14 = 44.1 - 14 =$ **30.1 kPa**
- (a) $z_c = 2c/(\gamma\sqrt{K_a}) = 20/(18\times 0.7) =$ **1.59 m**
- (b) <u>Without crack</u> (tension allowed) : $P_a = \frac{1}{2}\gamma H^2 K_a - 2cH\sqrt{K_a} = 110.25 - 70 =$ **40.3 kN/m**

  <u>With crack</u> (tension ignored) : $P_a = \frac{1}{2}\times 30.1\times(5-1.59) =$ **51.4 kN/m** ($= \frac{1}{2}\gamma H^2 K_a - 2cH\sqrt{K_a} + 2c^2/\gamma$).
- (c) $\bar{y} = (H - z_c)/3 = 3.41/3 =$ **1.14 m** above base.

Design with the crack (51.4 kN/m) – the safe value. If water fills the crack add $\frac{1}{2}\gamma_w z_c^2$.

10. Rankine vs Coulomb – Why Coulomb ?

- Rankine can't handle **wall friction**, rough/inclined wall backs, irregular backfill → **Coulomb's wedge theory** (1776) can.
- Rankine gives a state of stress (stress-field method, lower bound); Coulomb gives a failure-wedge method (limit equilibrium).

II. Coulomb's Wedge Theory (1776)

Assumptions

- Backfill is dry, **cohesionless**, homogeneous, isotropic; failure surface is a **plane** (BC) – a wedge ABC slides down.
- Wall **rough** : wall friction δ ($\approx$ ⅔φ) acts on back of wall; wall back can be inclined (angle α to horizontal); backfill can slope (β).
- Soil wedge behaves as a **rigid body**; failure is a plane-strain problem; for **active** case wedge moves down-and-out, for passive up-and-in.
- Pa is found by trial wedges and taking the **maximum** value (passive : **minimum**).

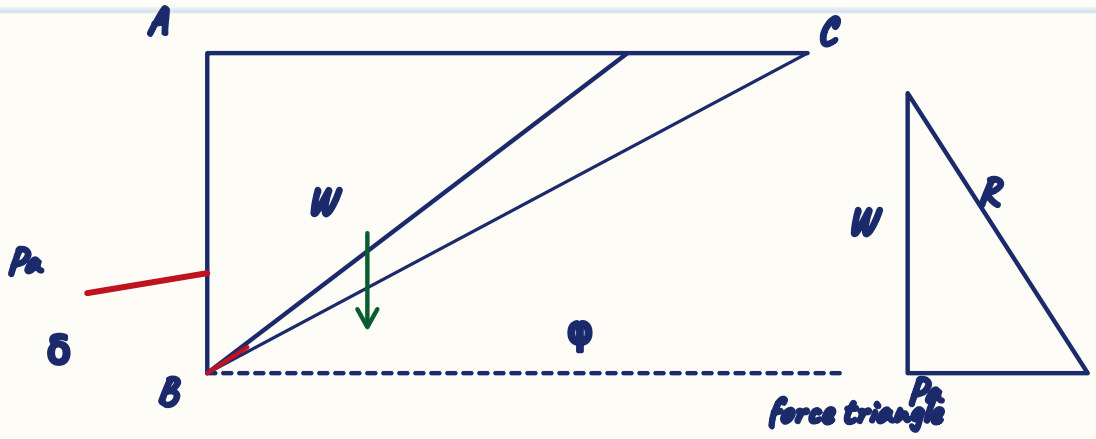


Fig 2 : Coulomb active wedge ABC and its force triangle (W, P_a, R)

- Forces on wedge: **W** (weight, known), P_a (reaction of wall, inclined δ to normal of wall), R (reaction on slip plane, inclined φ to normal). Triangle of forces closes → $P = W \sin(\theta-\varphi)/\sin(\ldots)$

$$P_a = \tfrac{1}{2} K_a \gamma H^2 \text{ (acts at } H/3 \text{ from base, at } \delta \text{ to wall normal)}$$

$$K_a = \sin^2(\alpha+\varphi) \,/\, \{ \sin^2\alpha \cdot \sin(\alpha-\delta) \cdot [1 + \sqrt{\,(\sin(\varphi+\delta)\sin(\varphi-\beta) / (\sin(\alpha-\delta)\sin(\alpha+\beta))\,}\,]^2 \}$$

- α = inclination of wall back with horizontal ; β = backfill slope ; δ = wall friction.
- **Special case** $\alpha = 90°, \delta = 0, \beta = 0 \rightarrow K_a = (1-\sin\varphi)/(1+\sin\varphi)$ = Rankine's K_a ✔.

Coulomb's passive pressure

$$P_p = \tfrac{1}{2} K_p \gamma H^2 \; ; \; K_p = \sin^2(\alpha-\varphi) / \{ \sin^2\alpha \cdot \sin(\alpha+\delta) \cdot [1 - \sqrt{(\sin(\varphi+\delta)\sin(\varphi+\beta) / (\sin(\alpha+\delta)\sin(\alpha+\beta))}\,]^2 \}$$

- For passive case P_p is the **minimum** over trial wedges ; with $\delta > \varphi/3$ the plane-surface assumption over-estimates P_p (unsafe) → use log-spiral surface or take $\delta = 0$ for passive.

Point of application : Coulomb gives only the magnitude; assume it acts at H/3 (for uniform γ, no surcharge). With surcharge → find C.G. of the pressure diagram.

Example 3 – Coulomb K_a

Vertical wall $H = 6$ m, backfill horizontal, $\varphi = 30°$, $\delta = 20°$, $\gamma = 18$ kN/m³. Find P_a.

- $\alpha = 90°$, $\beta = 0$: $\sin^2(120°) = 0.75$; $\sin(\alpha-\delta) = \sin 70° = 0.940$; $\sin(\varphi+\delta) = \sin 50° = 0.766$; $\sin\varphi = 0.5$
- $\sqrt{(0.766 \times 0.5 / 0.940)} = 0.638 \rightarrow [1.638]^2 = 2.684$
- $K_a = 0.75 / (0.940 \times 2.684) = \mathbf{0.297}$ (Rankine gives 0.333)

$$P_a = \tfrac{1}{2} \times 0.297 \times 18 \times 6^2 = 96.3 \text{ kN/m}$$

- $P_h = P_a \cos 20° = 90.5$ kN/m ; $P_v = P_a \sin 20° = 32.9$ kN/m (acts downward at $H/3 = 2$ m above base)

12. Rankine vs Coulomb – Quick Comparison

Rankine	Coulomb
Stress-state (plastic equilibrium) of whole backfill	Limit equilibrium of a rigid wedge
Wall smooth, vertical ($\delta = 0$)	Wall rough, any inclination, $\delta \neq 0$
Gives pressure **distribution** (p vs z)	Gives only total thrust P_a
Applicable to c–φ soil	Cohesionless soil ($c = 0$), dry
Passive – fairly accurate	Passive over-estimated if δ large
Inclined backfill : thrust ∥ ground surface	Thrust at δ to wall normal

13. Culmann's Graphical Method

➤ Graphical version of Coulomb's trial-wedge method – works for **any ground profile, surcharge, layered backfill, c = 0.** Gives P_a and the critical slip plane.

Steps (active case)

1. Draw wall back AB to scale (B = heel, A = top) and the ground line AD.
2. Draw **φ-line** (base line) BD' from B inclined at φ to the horizontal, up to meet the ground line.
3. Draw **pressure line** (ψ-line) from B making angle $\psi = \alpha - \delta$ with the base line (for smooth vertical wall $\psi = 90°$).
4. Choose trial slip planes BC_1, BC_2, BC_3 … ending on the ground; compute weight of each wedge W_1, W_2… (area × γ ; add surcharge load if any).
5. Along the base line from B, mark off W_1, W_2… to a convenient force scale (points 1, 2, 3 …).
6. Through points 1, 2, 3 draw lines **parallel to the pressure line** to meet respective trial planes BC_1, BC_2… at points c_1, c_2…
7. Join c_1, c_2… by a smooth curve = **Culmann's curve.**
8. Draw a **tangent to the curve parallel to the base line;** the point of tangency T gives the critical plane (join B–T and extend).
9. Draw TP ∥ pressure line to cut base line at P : P_a = **length TP × force scale.**

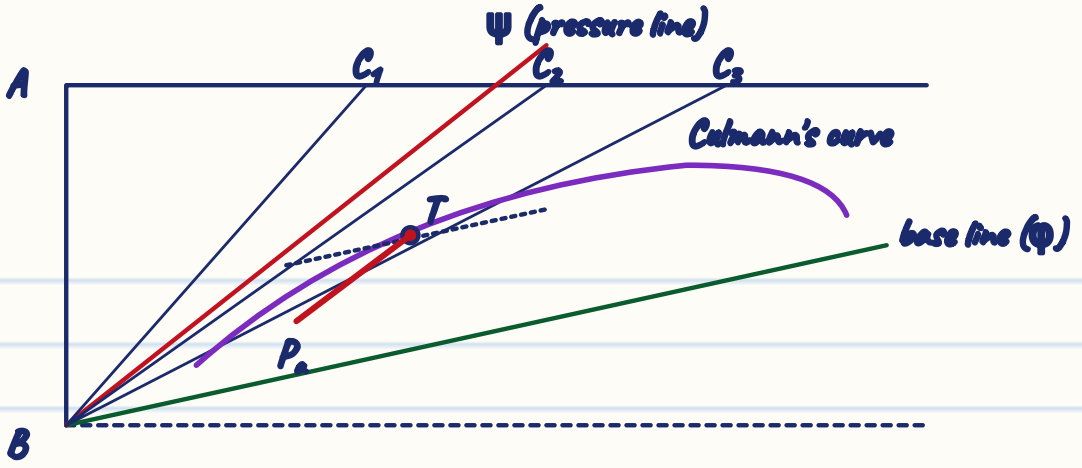


Fig 3 : Culmann's construction (schematic)

Passive case : same procedure but φ-line is drawn below, pressure line on the other side, and the **minimum** ordinate (tangent on the lower side of the curve) gives P_p.

14. Rebhann's (Poncelet's) Graphical Method

- A purely geometrical **construction for Coulomb's wedge** – the critical wedge and P_a are obtained directly, without trial wedges (quicker than Culmann). Only for a **plane ground surface** and uniform soil.

Steps (wall AB, A = top, B = base)

1. Draw wall back AB, ground line AD (slope β), **φ-line** BD from B (angle φ with horizontal) meeting ground line at D.
2. Draw the **pressure (ψ) line** through B as in Culmann ($\psi = \alpha - \delta$ with the φ-line).
3. Through A draw a line $\parallel$ ψ-line to meet BD at C.
4. Draw a **semicircle on BD** as diameter. Through C draw CE $\perp$ BD to meet the semicircle at E.
5. With B as centre and BE as radius cut BD at F.
6. Through F draw FG $\parallel$ ψ-line to meet the ground line AD at G. Join B–G : **BG = critical failure plane.**
7. With F as centre and FG as radius, cut BD at H. Join G–H to form **triangle GHF.**
8. Drop the perpendicular GI on BD (height x). Then

$$P_a = \gamma \times (\text{area of } \triangle GHF) = \tfrac{1}{2}\gamma \cdot HF \cdot x$$

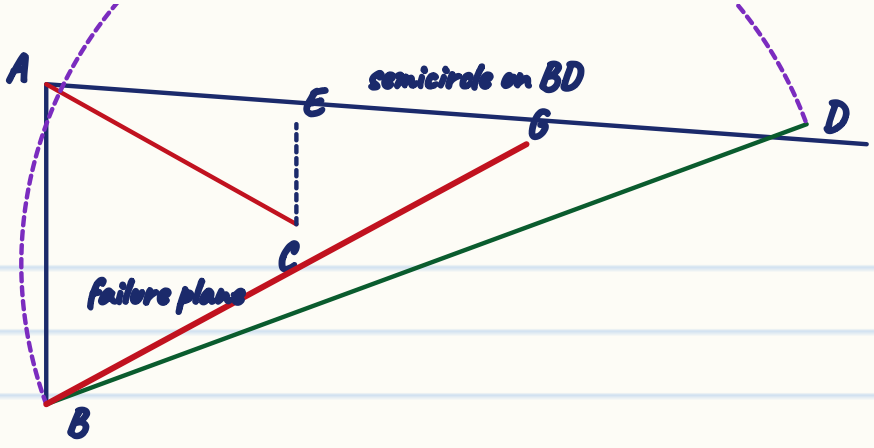


Fig 4 : Rebhann's construction (schematic – lettering as in the steps above)

Exam tip : write the 8 steps in order and draw a neat figure – marks are given for the construction sequence.

15. Active Pressure with Surcharge (Coulomb)

- **Uniform surcharge q** : replace by an equivalent height of soil $h_s = q/\gamma$; $P_a = \frac{1}{2} K_a \gamma H^2 + K_a q H$ (acts at C.G. of trapezoid) – same as Rankine.
- **Line load Q** at distance x : Culmann's method – add Q to W of every wedge that includes the load.

RETAINING WALLS

16. Types of Retaining Walls

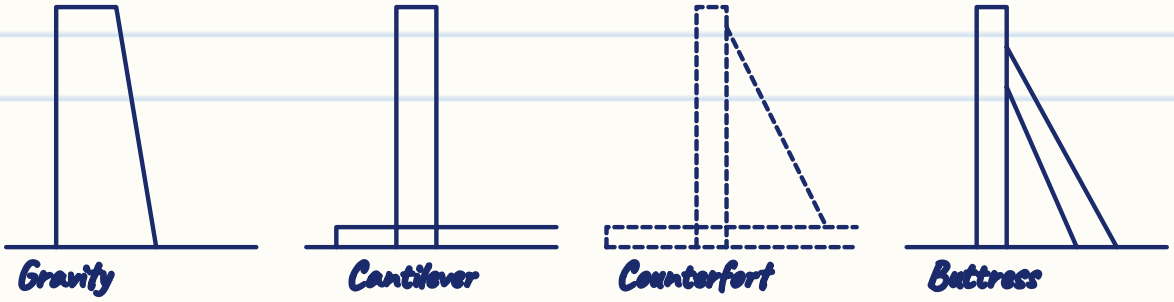


Fig 5 : Common types of retaining walls

Type	Action / Suitable height
Gravity wall (masonry / plain concrete)	Stability from own weight; no steel; H up to ≈ 3–4 m; base width 0.5–0.7 H, top ≥ 0.3 m
Semi-gravity	Gravity wall with light reinforcement near back face; slightly thinner
Cantilever (L / inverted T)	R.C.C. stem + base slab (heel + toe); weight of soil on heel helps stability ; H ≈ 3–7 m; base ≈ 0.4–0.7 H
Counterfort	Vertical ribs inside backfill (in tension) tie stem to heel slab; slab spans horizontally ; H > 6–7 m
Buttress	Ribs on the outer (toe) side, act in compression; more economical but uses space ; H > 6 m
Crib wall / Gabion wall	Interlocking boxes / wire cages filled with stone ; flexible, free-draining; tolerate settlement
Reinforced earth wall	Facing panels + steel / geosynthetic strips in compacted backfill (Henri Vidal) ; very high walls, flexible
Bridge abutment	Retains approach embankment + carries bridge loads
Sheet pile (bulkhead) wall	Cantilever / anchored ; for waterfronts, excavations

17. Drainage of Backfill

- Water adds hydrostatic thrust ($\frac{1}{2}\gamma_w h^2$) → can nearly double the load. Provide: **weep holes** (100 mm ø @ 2–3 m c/c), **filter / graded gravel** behind wall, longitudinal **perforated drain pipe**, surface slope / paving to keep rain out.
- Use **granular (free-draining) backfill** – clay backfill is avoided (swelling, high pressure).

18. Stability of Retaining Walls – Forces

- Forces per metre run : (i) **weight of wall W** (split into rectangles/triangles), (ii) **weight of soil over heel**, (iii) **active thrust P_a** (Rankine for cantilever – on a virtual vertical plane through heel, Coulomb for gravity), (iv) **passive resistance P_p** in front of toe (often neglected – may be scoured), (v) base reaction (N and friction F), (vi) surcharge, water pressure if any.

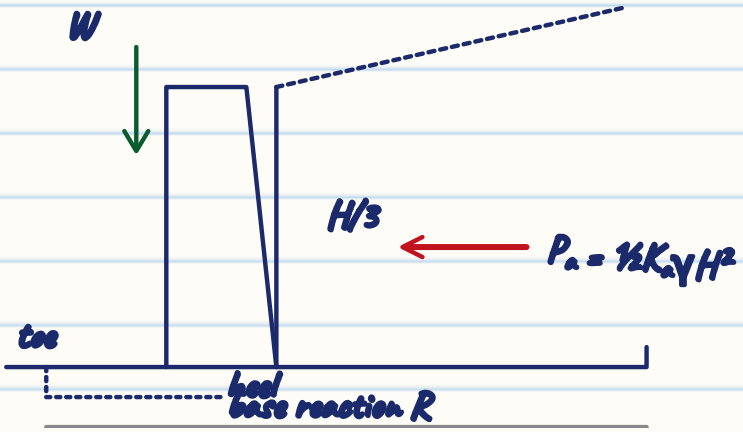


Fig 6 : Forces on a retaining wall

19. Stability Checks (per metre run)

(a) Overturning about toe

$$F.S._{overturning} = \Sigma M_R / \Sigma M_O = (W \cdot x + \ldots) / (P_h \cdot H/3) \geq 2.0$$

- ΣM_R = restoring moments of W, soil on heel (and P_v if any), about toe ; ΣM_O = overturning moment of horizontal thrust (and water). (IS 14458 : ≥ 1.5 with load factors; classical ≥ 2.)

(b) Sliding along base

$$F.S._{sliding} = (\mu \Sigma V + c_a B + P_p) / \Sigma H \geq 1.5$$

- $\mu = \tan \delta_b$ (≈ 0.5–0.6 for concrete on sand/gravel, $\tan \frac{2}{3}\varphi$ typically) ; c_a = adhesion (clay base). If F.S. < 1.5 → provide a **shear key** below base, increase base width, or tilt the base.

(c) Bearing pressure / tension at base

- Resultant ΣV acts at $\bar{x}$ from toe : $\bar{x} = (\Sigma M_R - \Sigma M_O) / \Sigma V$; eccentricity $e = B/2 - \bar{x}$

$$q_{max,min} = (\Sigma V / B)(1 \pm 6e/B) \text{ ; no tension if } e \leq B/6 \text{ (middle third)}$$

- $q_{max} \leq$ safe bearing capacity (F.S. ≥ 2.5–3 on q_{ult}) ; if $e > B/6$: $q_{max} = 2\Sigma V / (3\bar{x})$ (tension zone neglected).

(d) Other checks

- **Deep-seated (slope) stability** of the wall + soil mass (Unit II methods) ; F.S. ≥ 1.3–1.5.

- **Structural design** of stem, heel, toe slabs as cantilevers; for gravity walls check no tension in masonry at any section (for sections $e \leq b/6$).
- **Settlement** – tilt due to differential settlement (q_{max} at toe).

Proportioning of a cantilever wall (thumb rules)

- Base width $B \approx 0.5 - 0.7\ H$; base thickness $\approx H/10 - H/12$; stem thickness at base $\approx H/10 - H/12$, at top $\geq$ 200–300 mm; toe projection $\approx B/3$; embedment depth $\geq$ 1 m (below frost / scour).

Example 4 – Gravity Wall Stability

A masonry wall (γ = 22 kN/m^3) 4 m high, top width 1 m, base width 2.5 m, vertical back face, battered front. Backfill horizontal sand : γ = 18 kN/m^3, φ = 30°. Base friction μ = 0.5. Check stability (Rankine).

Item	Force (kN/m)	Arm from toe (m)	Moment (kN·m)
W_1 = 1 × 4 × 22 (rect.)	88	2.0	176
W_2 = ½ × 1.5 × 4 × 22 (tri.)	66	1.0	66
Σ	**154**		**242**

- K_a = 1/3 ; P_a = ½ × (1/3) × 18 × 4^2 = **48 kN/m** at $H/3$ = 1.33 m → M_0 = 48 × 1.333 = **64 kN·m**
- F.S. overturning = 242 / 64 = **3.78 > 2** ✔
- F.S. sliding = (0.5 × 154) / 48 = 77 / 48 = **1.60 > 1.5** ✔
- $\bar{x}$ = (242 – 64)/154 = 1.156 m ; e = 1.25 – 1.156 = **0.094 m < B/6 = 0.417** ✔ (no tension)
- q_{max} = (154/2.5)(1 + 6×0.094/2.5) = 61.6 × 1.226 = **75.5 kN/m²** ; q_{min} = 61.6 × 0.774 = **47.7 kN/m² ≤ SBC** ✔

Wall is safe against overturning, sliding and base tension; bearing OK if SBC $\geq \approx$ 75 kN/m² (checked against allowable).

Example 5 – Passive Resistance & Sliding with Key

A wall of ΣV = 200 kN/m and ΣH = 110 kN/m rests on sand with μ = 0.45. Find F.S. against sliding without and with a 0.8 m deep key having passive resistance $P_p = \frac{1}{2} K_p \gamma h^2$ (K_p = 3, γ = 18).

- Without key : F.S. = 0.45 × 200 / 110 = 90/110 = 0.82 < 1.5 ✘ unsafe
- P_p (key) = ½ × 3 × 18 × 0.8^2 = 17.3 kN/m (use ⅔ of it for safety ≈ 11.5)
- With key : F.S. = (90 + 17.3)/110 = 0.98 – still < 1.5 → **increase base width / add key / heel extension** ; each extra metre of heel adds soil weight (↑ ΣV).

Important Questions – Unit III

1. Explain active, passive and at-rest earth pressure. Give the order of K_a, K_0, K_p. (2/5 M)
2. State the assumptions of Rankine's theory. Derive K_a and K_p for cohesionless soil. (10 M)
3. Derive active pressure for c–φ soil; find depth of tension crack and critical height. (10 M)
4. Draw the pressure diagram for a layered backfill with water table and surcharge; find total thrust and point of application. (10 M)
5. Derive Coulomb's active earth pressure for a cohesionless backfill; show it reduces to Rankine. (10 M)
6. Explain Culmann's graphical method with a neat sketch. (5–10 M)
7. Explain Rebhann's graphical method for finding active earth pressure. (5–10 M)
8. Compare Rankine and Coulomb theories. (5 M)
9. Describe the different types of retaining walls with sketches. (10 M)
10. Explain the stability checks for a gravity / cantilever retaining wall. Numerical on overturning, sliding, base pressure. (10 M)

Formula sheet : $K_a = \tan^2(45^\circ - \varphi/2)$; $K_p = 1/K_a$; $z_0 = 2c/(\gamma\sqrt{K_a})$; $H_c = 4c/(\gamma\sqrt{K_a})$; $P_a = \frac{1}{2}K_a\gamma H^2$

All the best! Revise the figures – they carry marks.

UNIT – IV : SHALLOW FOUNDATIONS

1. Foundation – Meaning & Functions

- **Foundation** = the lowest part of a structure which transmits the load of the superstructure safely to the soil / rock below.
- Two basic requirements of every foundation:
  - Safety against shear failure of soil (bearing capacity, F.S. ≥ 2.5 to 3)
  - Serviceability – total & differential settlement within permissible limits

2. Classification of Foundations

- **Shallow foundation** : $D_f \leq B$ (Terzaghi); load is carried mainly by base resistance. Eg: spread, strip, combined footing, raft.
- **Deep foundation** : $D_f > B$ ($\geq$ 3 to 4 B); load carried by skin friction + end bearing. Eg: piles, piers, wells (Unit V).
- Here D_f = depth of foundation, B = least lateral dimension of base, L = length.

3. Types of Shallow Foundations

(a) Isolated (Spread / Pad) Footing

- Under a single column; plan square, rectangular or circular. Most common and cheapest type.
- Types of section: plain concrete (block), stepped, sloped (tapered) and pedestal footing; R.C.C. for heavy loads.
- Used when soil is of good bearing capacity (S.B.C $\geq$ 150 kN/m^2) and columns are far apart so that adjacent footings do not overlap.

(b) Strip / Wall Footing

- Continuous footing under a load-bearing wall ($L \gg B$, plane-strain condition). Also used for a row of closely spaced columns.
- Brick/stone masonry walls: stepped (offset) footing with offset $\leq$ ½ of the step height; spread $\approx$ 2 × wall thickness.

(c) Combined Footing

- Supports **two or more columns**. Provided when (i) columns are so close that isolated footings overlap, (ii) a column is near the property line and the footing cannot project beyond it, (iii) bearing capacity is low.

- Rectangular combined footing : when loads of both columns are nearly equal (or the column with larger load has more space)
- Trapezoidal combined footing : when loads are unequal; centroid of base area is made to coincide with the resultant of column loads so pressure is uniform.

(d) Strap (Cantilever) Footing

- Two isolated footings joined by a **strap beam**; strap carries the moment due to eccentric load of the exterior column so that the footings behave independently. Economical when distance between columns is large (a combined footing would be heavy).

(e) Raft (Mat) Foundation

- A large continuous slab covering the whole area under the structure and supporting all columns / walls.
- Used when : (i) soil is weak, S.B.C. low – isolated footings would cover > 50 % of the plan area; (ii) differential settlement must be reduced; (iii) compressible pockets / erratic soil; (iv) basement & water-proofing needed; (v) heavy loads.
- Types: flat plate raft, raft with beams (slab-and-beam), raft with pedestals, cellular raft, piled raft.
- **Floating (compensated) raft:** weight of soil excavated ≈ weight of the structure, so net pressure on soil is almost zero → no settlement problem. Net pressure $q_n = q - \gamma D_f$

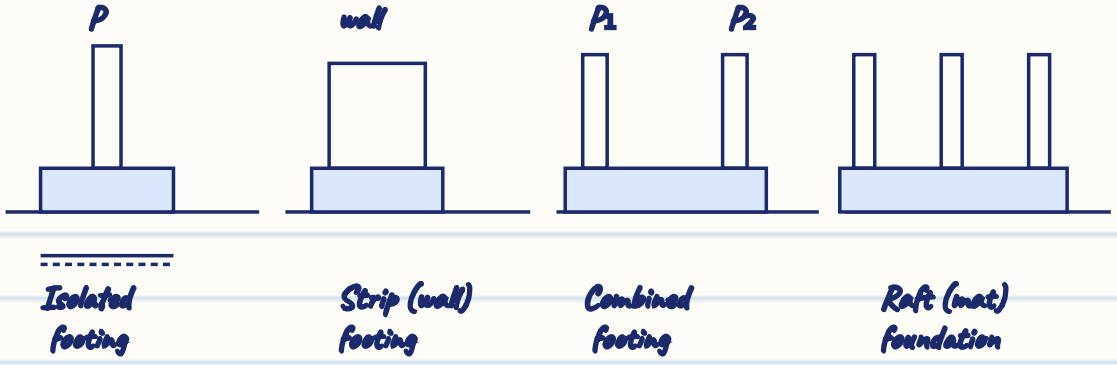


Fig. 1 – Common shallow foundations (side elevations)

5. Choice of Type of Foundation

Selection depends on a combination of factors – decided after soil investigation:

1. **Soil conditions** – type, S.B.C., compressibility, water table, presence of expansive / collapsible / filled soil.

2. **Magnitude & nature of loads** – dead, live, wind, seismic, eccentric / moment loads, dynamic loads.
3. **Type of superstructure** – flexible / rigid frame, sensitivity to differential settlement (framed building vs. tank).
4. **Depth of water table** & ground-water chemistry (sulphates, chlorides).
5. **Adjacent structures** – undermining, interference of stress bulbs, property line limits.
6. **Economy, construction time, available equipment**, local practice.

6. Depth & Location of Foundation

Depth D_f is governed by:

- **Bearing stratum** – founded below top soil / organic matter / filled ground, on a firm stratum. IS 1904: minimum depth 0.5 m below ground level.
- **Frost action** – below the frost line (in cold regions 0.6–1.5 m) to avoid heave.
- **Seasonal volume change** – in expansive (black cotton) soils go below the active zone ($\approx$ 1.5 – 3.5 m), or use under-reamed piles / sand cushion.
- **Scour** (bridges) – foundation at $\geq$ 1.2–2 m below maximum scour depth.
- **Water table** – above W.T. preferably, to avoid dewatering and buoyancy.
- **Adjacent foundations** – new footing should not be deeper than neighbour's unless it is at a distance; line of 2H : 1V (or 45°) from the old footing edge must be respected.
- **Underground facilities**, basements, lift pits, machine foundations.

Rankine's minimum depth (for granular soils)

$$D_f = (p / \gamma) \cdot [(1 - \sin\varphi) / (1 + \sin\varphi)]^2$$

- p = intensity of gross loading (kN/m^2), γ = unit weight of soil, φ = angle of shearing resistance. Gives the depth below which soil cannot be squeezed out laterally.

Location (plan): centroid of base must coincide with line of action of resultant load ($e \leq B/6$); keep clear of existing structures & slopes.

8. Bearing Capacity – Definitions

- **Ultimate bearing capacity (q_u)** : minimum gross pressure intensity at the base of the foundation at which the soil fails in shear.

- **Net ultimate bearing capacity (q_{nu})** : net increase in pressure over overburden : $q_{nu} = q_u - \gamma D_f$
- **Net safe bearing capacity (q_{ns})** : $q_{ns} = q_{nu} / F$ (F = factor of safety, usually 2.5 – 3)
- **Safe bearing capacity (q_s)** : $q_s = q_{ns} + \gamma D_f$
- **Gross safe bearing capacity** = q_u / F.
- **Safe / Allowable bearing pressure (q_a)** : the maximum net pressure which the soil can carry without shear failure AND without exceeding permissible settlement ⇒ q_a = smaller of (q_{ns}, pressure for tolerable settlement).

9. Terzaghi's Bearing Capacity Theory (1943)

Assumptions

- Footing is **strip (continuous)**, base is **rough**, soil is homogeneous, isotropic, **c–φ** soil.
- Shallow foundation : $D_f \leq B$; soil above base level replaced by uniform surcharge $q = \gamma D_f$ (its shear strength is neglected).
- Failure is by **general shear**; the Mohr–Coulomb criterion holds; load is vertical and concentric.

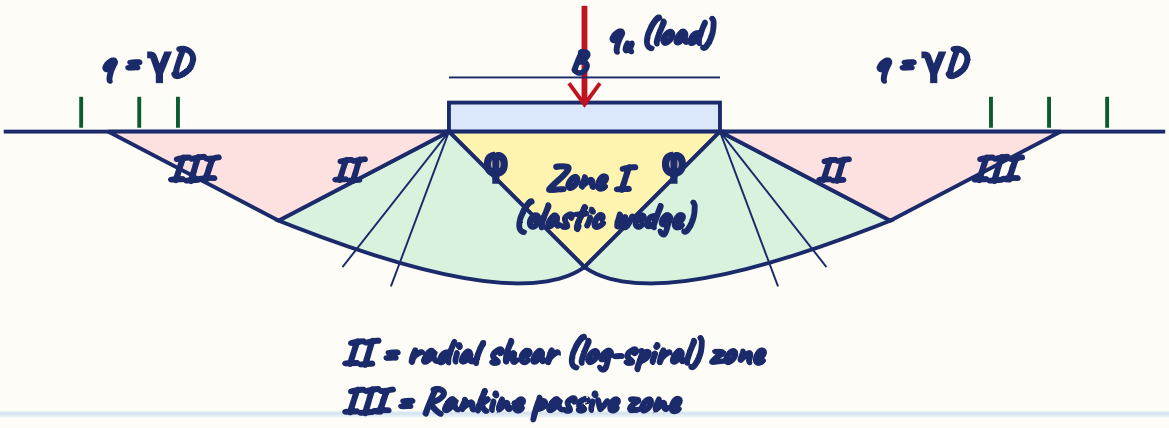


Fig. 2 – Terzaghi's failure mechanism (zones I, II, III) – base angles of wedge = φ

- **Zone I** : wedge below footing; moves down with footing; base angles = φ (rough base).
- **Zone II** : radial shear zone – curved boundary is a logarithmic spiral.
- **Zone III** : Rankine passive zones, base angles ($45° - \varphi/2$).

Outline of derivation (strip footing)

- Consider equilibrium of wedge I (per unit length). Forces: ① **downward** – $q_u \cdot B$ (load) + weight $W = \tfrac{1}{4}\gamma B^2 \tan\varphi$; ② **upward** – cohesion along two inclined faces = $2 \cdot C \cdot \sin\varphi$ where $C = c \cdot (B/2)/\cos\varphi$ per face ; and ③ passive reaction P_p on each face, inclined at φ to the normal.
- Vertical equilibrium $\Rightarrow q_u B + W - 2P_p - cB\tan\varphi = 0$
- P_p is made up of three parts: P_{pc} (due to cohesion c), P_{pq} (due to surcharge q) and $P_{p\gamma}$ (due to unit weight γ) $\Rightarrow$ solved with the log-spiral $\Rightarrow$ superposition gives :

$$q_u = c\,N_c + \gamma\,D_f\,N_q + \tfrac{1}{2}\gamma\,B\,N_\gamma$$

(strip footing, general shear failure)

- The 3 terms : cohesion component + surcharge component + self-weight (friction) component.
- N_c, N_q, N_γ = Terzaghi's bearing capacity factors – functions of φ only.

Shape Factors (Terzaghi)

Footing	q_u (general shear)
Strip	$cN_c + qN_q + 0.5\gamma BN_\gamma$
Square (B × B)	$1.3cN_c + qN_q + 0.4\gamma BN_\gamma$
Circular (dia B)	$1.3cN_c + qN_q + 0.3\gamma BN_\gamma$
Rectangular (B × L)	$cN_c(1+0.3B/L) + qN_q + 0.5\gamma BN_\gamma(1-0.2B/L)$

General vs. Local Shear Failure

- **General shear** (dense sand, stiff clay) : well-defined failure surface reaching ground; sudden failure, bulging, clear peak on the load–settlement curve.
- **Local shear** (loose sand, soft clay) : failure surface not continuous to the surface; gradual failure, no sharp peak – settlement is large.
- For local shear use reduced strength (Terzaghi) :

$$c' = (2/3)\,c \qquad \tan\varphi' = (2/3)\tan\varphi$$

- ...and use N'_c, N'_q, N'_γ (values at φ'). Eg. strip: $q'_u = (2/3)c\,N'_c + qN'_q + \tfrac{1}{2}\gamma BN'_\gamma$. Square: $q'_u = 0.867cN'_c + qN'_q + 0.4\gamma BN'_\gamma$.

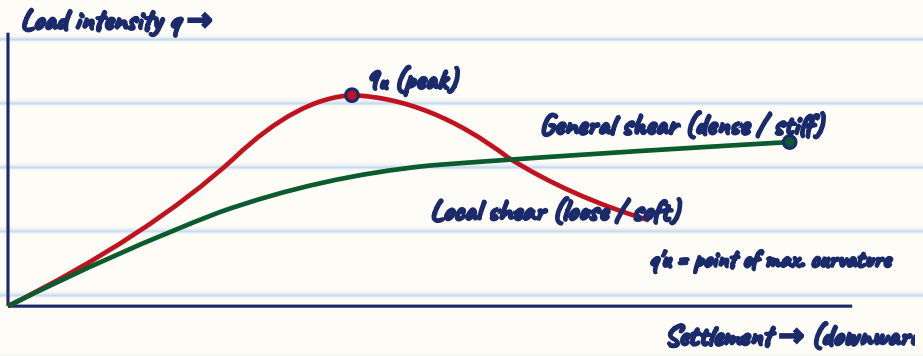


Fig. 3 – Load–settlement curves : general and local shear (axes drawn schematically)

Effect of Water Table (Terzaghi)

Submerged soil has $\gamma' = \gamma_{sat} - \gamma_w$ ($\approx \tfrac{1}{2}\gamma_{sat}$) → bearing capacity reduces. Let D_w = depth of W.T. below ground level.

- **Case 1 : W.T. above base level ($D_w \le D_f$)**
  - $q = \gamma D_w + \gamma'(D_f - D_w)$; use γ' in the 3rd term ; ($\gamma_w = 9.81\ kN/m^3$)
- **Case 2 : W.T. at base level ($D_w = D_f$)**
  - $q = \gamma D_f$ (no change) ; γ' in 3rd term.
- **Case 3 : W.T. below base but within B ($D_f < D_w < D_f + B$)**
  - $q = \gamma D_f$; γ in 3rd term replaced by average : $\boldsymbol{\gamma_{av} = \gamma' + (D_w - D_f)(\gamma - \gamma')/B}$
- **Case 4 : $D_w \ge D_f + B \Rightarrow$ no effect of W.T.**

Reduction-factor (IS 6403) method

$$q_u = cN_c + qN_q W' + \tfrac{1}{2}\gamma B N_\gamma W''$$

- $W' = 0.5\,[1 + D_{w1}/D_f]$ (W.T. depth D_{w1} measured from ground; $W' = 0.5$ if W.T. at ground level, 1 if at base or below).
- $W'' = 0.5\,[1 + D_{w2}/B]$ (D_{w2} = depth of W.T. below the base; $W'' = 0.5$ at base, 1 if $D_{w2} \ge B$).

Rise of water table to the base reduces q_u of a cohesionless soil by about 50 %. Always design for the worst (highest) W.T.

Terzaghi's Bearing Capacity Factors

$\varphi°$	0	5	10	15	20	25	30	35	40
N_c	5.7	7.3	9.6	12.9	17.7	25.1	37.2	57.8	95.7
N_q	1.0	1.6	2.7	4.4	7.4	12.7	22.5	41.4	81.3
N_γ	0	0.5	1.2	2.5	5.0	9.7	19.7	42.4	100.4

Worked Example 1 – Terzaghi (strip footing, c–φ soil)

Q. A strip footing 2 m wide is founded at 1.5 m depth in a soil with $c = 20\ kN/m^2$, $\varphi = 20°$, $\gamma = 18\ kN/m^3$. Find the safe bearing capacity for $F = 3$ (general shear). Take $N_c = 17.7$, $N_q = 7.4$, $N_\gamma = 5.0$.

Sol.

- $q = \gamma D_f = 18 \times 1.5 = 27\ kN/m^2$
- $q_u = cN_c + qN_q + \tfrac{1}{2}\gamma BN_\gamma$
- $= 20(17.7) + 27(7.4) + 0.5(18)(2)(5.0)$
- $= 354 + 199.8 + 90 = 643.8\ kN/m^2$
- $q_{nu} = q_u - \gamma D_f = 643.8 - 27 = 616.8\ kN/m^2$
- $q_{ns} = 616.8 / 3 = 205.6\ kN/m^2$

$$q_s = q_{ns} + \gamma D_f = 205.6 + 27 \approx 232.6\ kN/m^2$$

Worked Example 2 – Square footing with water table

Q. A square footing 2.5 m × 2.5 m is placed at 1.5 m depth in sand with $c = 0$, $\varphi = 30°$, $\gamma = 18\ kN/m^3$ above W.T. and $\gamma_{sat} = 20\ kN/m^3$. W.T. is at 1 m below ground level. Find the net safe bearing capacity ($F = 3$) and the safe load. ($N_q = 22.5$, $N_\gamma = 19.7$.)

Sol. W.T. above base ⇒ Case 1 ($D_w = 1\ m < D_f$).

- $\gamma' = 20 - 9.81 = 10.19\ kN/m^3$
- $q = \gamma D_w + \gamma'(D_f - D_w) = 18(1) + 10.19(0.5) = 23.1\ kN/m^2$
- Square footing, $c = 0$: $q_u = qN_q + 0.4\gamma' BN_\gamma$
- $= 23.1(22.5) + 0.4(10.19)(2.5)(19.7)$
- $= 519.6 + 200.7 = 720.4\ kN/m^2$
- $q_{nu} = 720.4 - 23.1 = 697.3\ kN/m^2 \Rightarrow q_{ns} = 697.3/3 = 232.4\ kN/m^2$

- Safe load = $q_{ns} \times B^2 = 232.4 \times 6.25 \approx$ **1452 kN** (net) [gross: $q_s = 255.5 \rightarrow$ 1597 kN]

Had the W.T. been at the base or deeper than B below it, qu would have been ≈ 40–50 % higher – it shows how important the water table is!

10. Meyerhof's Bearing Capacity Theory (1951, 1963)

- Improves Terzaghi : considers **shear strength of soil above the base** (failure surface extends to ground), shape, depth and load-inclination effects. Footing may be rough or smooth.
- Failure zones : elastic wedge + radial shear + mixed shear zone (a log-spiral surface rising to the ground surface – equivalent free surface).

General equation

$$q_u = c\,N_c\,s_c\,d_c\,i_c + q\,N_q\,s_q\,d_q\,i_q + \tfrac{1}{2}\gamma\,B\,N_\gamma\,s_\gamma\,d_\gamma\,i_\gamma$$

s = shape factors, d = depth factors, i = inclination factors. (For strip footing s = 1.) $q = \gamma D_f$

Bearing capacity factors (Meyerhof)

$$N_q = e^{\pi \tan\varphi}\tan^2(45^\circ + \varphi/2)\,;\; N_c = (N_q - 1)\cot\varphi\,;\; N_\gamma = (N_q - 1)\tan(1.4\varphi)$$

φ°	0	5	10	15	20	25	30	35	40
N_c	5.14	6.49	8.35	10.98	14.83	20.72	30.14	46.12	75.31
N_q	1.0	1.57	2.47	3.94	6.40	10.66	18.40	33.30	64.20
N_γ	0	0.45	1.22	2.65	5.39	10.88	22.40	48.03	109.4

Shape, depth & inclination factors

Let $K_p = \tan^2(45^\circ + \varphi/2)$ (coefficient of passive pressure), α = inclination of load with the vertical.

Factor	$\varphi = 0$	$\varphi > 10°$
Shape s_c	$1 + 0.2\,(B/L)\,K_p$	
$s_q = s_\gamma$	1	$1 + 0.1\,(B/L)\,K_p$
Depth d_c	$1 + 0.2\,(D_f/B)\,\sqrt{K_p}$	
$d_q = d_\gamma$	1	$1 + 0.1\,(D_f/B)\,\sqrt{K_p}$
Inclination $i_c = i_q$	$(1 - \alpha°/90°)^2$	
i_γ	0 (for $\varphi = 0$)	$(1 - \alpha°/\varphi°)^2$

For $\varphi = 0$ clay : $s_c = 1 + 0.2B/L$. For strip ($B/L = 0$) all shape factors = 1; for vertical load $i = 1$.

II. Skempton's Method (for Clays, $\varphi_u = 0$)

- For **saturated clays under undrained loading**; $c = c_u$, $\varphi = 0 \Rightarrow N_q = 1$, $N_\gamma = 0$.
- Skempton (1951) showed that N_c is not constant – it increases with the depth/width ratio D_f/B (because of the increase of shear resistance of the soil above base level) and with footing shape.

$$q_{nu} = c_u N_c \quad \Rightarrow \quad q_u = c_u N_c + \gamma D_f$$

Skempton's N_c values

- **Strip** : $N_c = 5\,(1 + 0.2\,D_f/B)$ for $D_f/B \leq 2.5$; $N_c = 7.5$ for $D_f/B > 2.5$
- **Square / circular** : $N_c = 6\,(1 + 0.2\,D_f/B)$ for $D_f/B \leq 2.5$; $N_c = 9$ for $D_f/B > 2.5$
- **Rectangular** ($B \times L$) : N_c (rect) = N_c (strip) · $(0.84 + 0.16\,B/L)$

D_f/B	0	0.5	1.0	1.5	2.0	2.5	≥ 4
N_c strip	5.0	5.5	6.0	6.5	7.0	7.5	7.5
N_c square	6.0	6.6	7.2	7.8	8.4	9.0	9.0

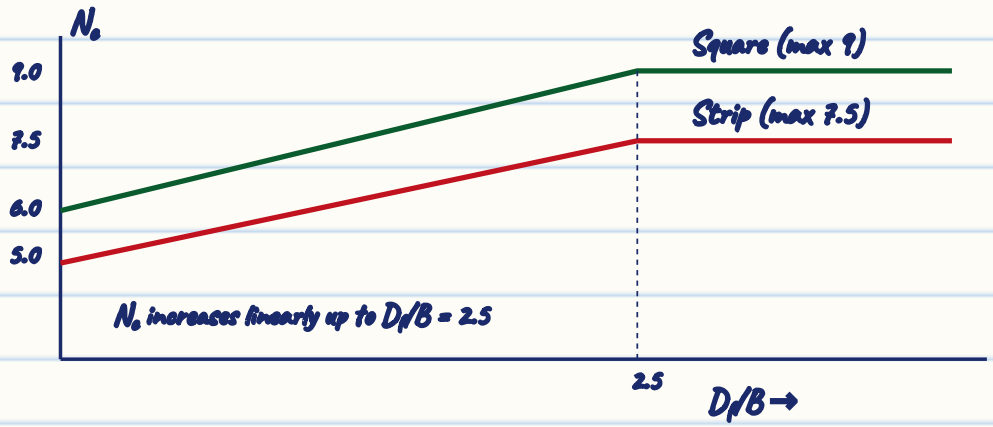


Fig. 4 – Skempton's N_c vs D_f/B (schematic)

Net safe bearing capacity

$$q_{ns} = c_u N_c / F \;;\; q_s = q_{ns} + \gamma D_f$$

- F = 2.5 to 3. For a surface footing on clay (strip) : $q_{nu} = 5.14\, c_u$ (Prandtl / Meyerhof) ≈ $5\, c_u$ (Skempton).
- Eg: c_u = 60 kN/m², square footing B = 2 m, D_f = 1 m, γ = 18 : D_f/B = 0.5 ⇒ N_c = 6.6; q_{nu} = 396; q_{ns} = 132; q_s = 132 + 18 = 150 kN/m².

ALLOWABLE BEARING PRESSURE

12. Safe Bearing Pressure from N-value (SPT)

- SPT (IS 2131) gives N = no. of blows for 300 mm penetration of a split spoon – a quick, in-situ indicator of density (sands) & consistency (clays).
- In sands, settlement (not shear) usually governs ⇒ allowable pressure is based on **25 mm** (or 40 mm) of settlement.

Corrections to field N (IS 2131)

- **Overburden** : $N_c = C_N N$, $C_N = 0.77 \log_{10}(2000/\sigma'_v)$ (σ'_v in kN/m², for $\sigma'_v \geq 25$)
- **Dilatancy** (saturated fine / silty sand, $N > 15$) : $N_c = 15 + ½ (N - 15)$
- Also : energy, rod length, borehole & sampler corrections in N_{60}.

Sands			Clays		
N	Density	φ°	N	Consistency	q_u kN/m²
0–4	V. loose	< 29	< 2	V. soft	< 25
4–10	Loose	29–30	2–4	Soft	25–50
10–30	Medium	30–36	4–8	Medium	50–100
30–50	Dense	36–41	8–15	Stiff	100–200
> 50	V. dense	> 41	15–30	V. stiff	200–400

Peck's approximation : $\varphi \approx 27 + 0.3\,N$ (degrees) ; for clays $c_u = q_u/2 \approx 6N$ to $7N$ (kN/m²)

(i) Terzaghi–Peck method

- Based on charts of allowable pressure vs. footing width B for different N (for 25 mm settlement). Footing on sand : q_a increases with B for shear but decreases for settlement ⇒ for B > ~1.2 m, q_a becomes nearly constant.
- Water table correction : if W.T. is at / above base, multiply q_a by **0.5** (W.T. at base ≈ 0.5 – 0.6 effect).

(ii) Meyerhof's equations (25 mm settlement, N_{60})

$$B \le 1.2\ \text{m}: \quad q_a\,(\text{net}) = 12\,N\,K_d \ \text{kN/m}^2$$
$$B > 1.2\ \text{m}: \quad q_a\,(\text{net}) = 8\,N\,[(B + 0.3)/B]^2\,K_d \ \text{kN/m}^2$$

- $K_d = 1 + 0.33\,D_f/B \le 1.33$ (depth factor). N = average SPT value over depth B below base (corrected). Raise by 50 % for 40 mm settlement. Halve the value if W.T. is at the base level.

13. Plate Load Test (IS 1888 – 1982)

- A **field test** to obtain ultimate bearing capacity and settlement of soil at the foundation level by loading a rigid plate (model footing). Test is done in a pit of size ≥ 5 × plate width at the proposed level.
- Plate : mild steel, 25 mm thick, **square 300–750 mm** or circular of 300 – 750 mm dia. (usually 300 mm for sand; 450 – 600 for clay).

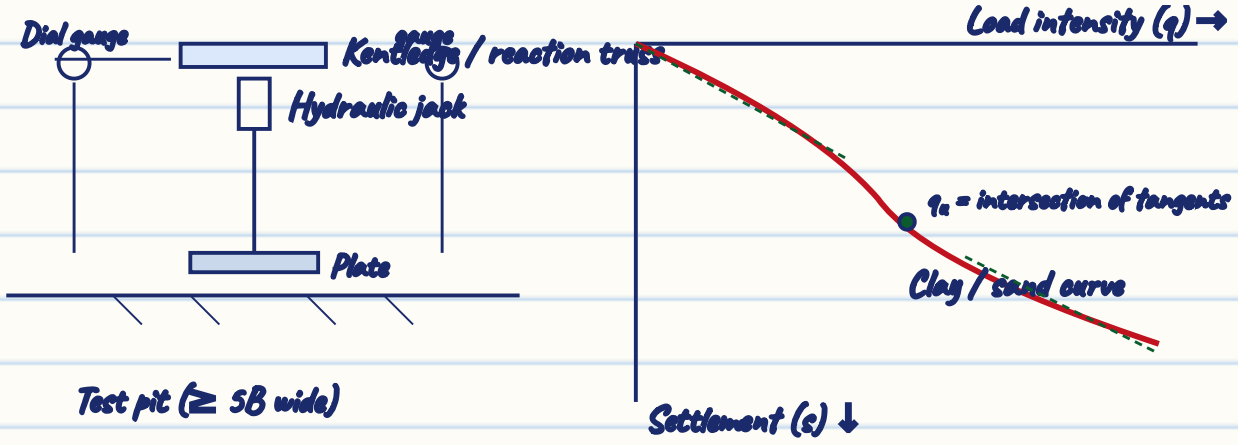


Fig. 5 – Plate load test arrangement and load–settlement curve

Procedure

1. Excavate pit, level the bottom; place plate on thin sand bed, seat it with a seating load of 7 kN/m² and release.
2. Apply load in increments of ≈ 1/5 of the estimated safe load (or 1/10 of ultimate), through a hydraulic jack reacting against kentledge / anchored truss.
3. Record settlement by 2–4 dial gauges (0.01 mm) at time intervals 1, 2.25, 4, 6.25, 9, 16, 25 … min; next load applied when rate of settlement < 0.02 mm/min (or after 1 hr).
4. Continue till failure / total settlement of 25 mm (or 1/10 B) or twice the design intensity.

Interpretation

- Plot load intensity q vs. settlement s (arithmetic and log–log). q_u is at the break point (intersection of tangents). If no break : q at s = 25 mm / 10 % of B_p.
- **Bearing capacity (size effect):** Clay : $q_f = q_p$ (independent of size) ; Sand : $q_f = q_p \cdot (B_f/B_p)$
- Safe B.C. = q_u/F ; and settlement governs.

Settlement of footing from plate settlement ($S_p \rightarrow S_f$)

$$\text{Sand}: \quad S_f = S_p \cdot [\, B_f (B_p + 0.3) \,/\, (\, B_p (B_f + 0.3)\,)\,]^2$$

$$\text{Clay}: \quad S_f = S_p \cdot (B_f / B_p)$$

(B in metres; same intensity of loading q for plate and footing.)

Example 4. A 300 mm square plate settles 12 mm under a certain pressure on sand. Find the settlement of a 2 m square footing at the same pressure.

- $S_f = 12 \times [2\,(0.3 + 0.3) / (0.3\,(2 + 0.3))]^2 = 12 \times (1.2 / 0.69)^2 = 12 \times 3.025 \approx$ 36.3 mm

Limitations of plate load test

- Plate stress bulb $\approx 2B_p$ deep – does **not** test deeper layers (weak layer below remains undetected).
- Does not give long-term consolidation settlement in clays (test lasts hours).
- Scale effect; ground-water level effect; costly & time-consuming.

14. Allowable (Permissible) Settlements

- **Total settlement (S_t)** = immediate (S_i) + primary consolidation (S_c) + secondary compression (S_s).
- **Differential settlement (δ)** : difference in settlement between two points of the same structure; **angular distortion (β)** = δ / L (L = distance between the points).
- Differential settlement – causes cracks, tilting, distress; more serious than uniform settlement. Uniform settlement (even 100 mm) is usually harmless except for connecting services.
- Limits depend on structure type, rigidity, soil, and use. Sands : $S_{diff} \approx 0.5$–$0.75\ S_{max}$; clays : $S_{diff} \approx 0.5\ S_{max}$ (Skempton & MacDonald).

IS 1904 – 1986 permissible settlements

Foundation	Soil	Max. S (mm)	Differential	Ang. dist.
Isolated footing	Sand / hard clay	50	0.0033 L	1/300
	Plastic clay	75	0.0033 L	1/300
Raft	Sand / hard clay	75	0.0033 L	1/300
	Plastic clay	100	0.0033 L	1/300

Angular distortion limits (Bjerrum, 1963)

$\beta = \delta/L$	Effect
1/750	Limit for machinery sensitive to settlement
1/500	Safe limit for framed buildings with no cracking
1/300	Cracking of panel / partition walls begins
1/250	Tilting of tall rigid buildings visible
1/150	Structural damage to general buildings

PILE & WELL FOUNDATIONS

1. Pile Foundation – Introduction

Pile = slender structural member (timber / concrete / steel) that transfers load to deeper, stronger strata by skin friction + end bearing.

When are piles used?

- Top soil weak / compressible, firm stratum at depth (> 3–4 m)
- Heavy concentrated loads – bridges, towers, silos, jetties
- Uplift & lateral loads (chimneys, transmission towers, quay walls)
- Scour-prone river beds, expansive / collapsible soils
- High water table – deep excavation uneconomical

2. Classification of Piles

(a) Based on function / load transfer

- **End-bearing (point-bearing) pile** – tip rests on rock / hard stratum; load mostly by tip: $Q_u \approx Q_p$
- **Friction pile** – floats in soft clay / loose sand; load by shaft: $Q_u \approx Q_s$
- **Combined** – $Q_u = Q_p + Q_s$ (most common)
- **Compaction pile** – densify loose granular soil
- **Tension / uplift pile** – resist upward pull (towers, basements)
- **Batter (raker) pile** – inclined; resist horizontal loads
- **Sheet pile** – retaining walls / cofferdams
- **Anchor pile** – hold down bulkheads; **Fender pile** – protect jetties

(b) Based on material

- Timber, Concrete (precast / cast-in-situ / prestressed), Steel (H, pipe), Composite

(c) Based on method of installation

- **Driven** (displacement) – hammer / vibro; **Bored** (non-displacement); **Driven cast-in-situ**; **Screw piles**; **Jacked**

(d) Based on size

- Small dia. < 600 mm; Large dia. ≥ 600 mm (bored piers / caissons)

3. Types of Piles – Material wise

Type	Features / Uses	Merits & Demerits
Timber	Sal, teak, deodar; 6–12 m long, 0.2–0.5 m dia. Temporary, marine fenders, light loads (≤ 250 kN)	Cheap, light, easy handling. Rots above W.T.; marine borer attack; low capacity
Precast RC	Square / octagonal; cast in yard, cured, then driven. Reinforced for handling. 300–600 mm; ≤ 30 m	Durable, good quality control. Heavy, hard to cut / extend; need driving equipment
Prestressed	Long, slender; resists driving tension & bending	Light, high capacity, less cracking; costly
Cast-in-situ	Bored / driven with casing – Franki, Raymond, Vibro, Under-reamed	Any length, no handling stress. Necking / segregation risk
Steel	H-pile, pipe pile (filled with concrete). Easy to splice	High capacity, deep penetration. Corrosion, costly
Composite	Timber + concrete / steel + concrete (upper part durable)	Economical for long piles with high W.T.

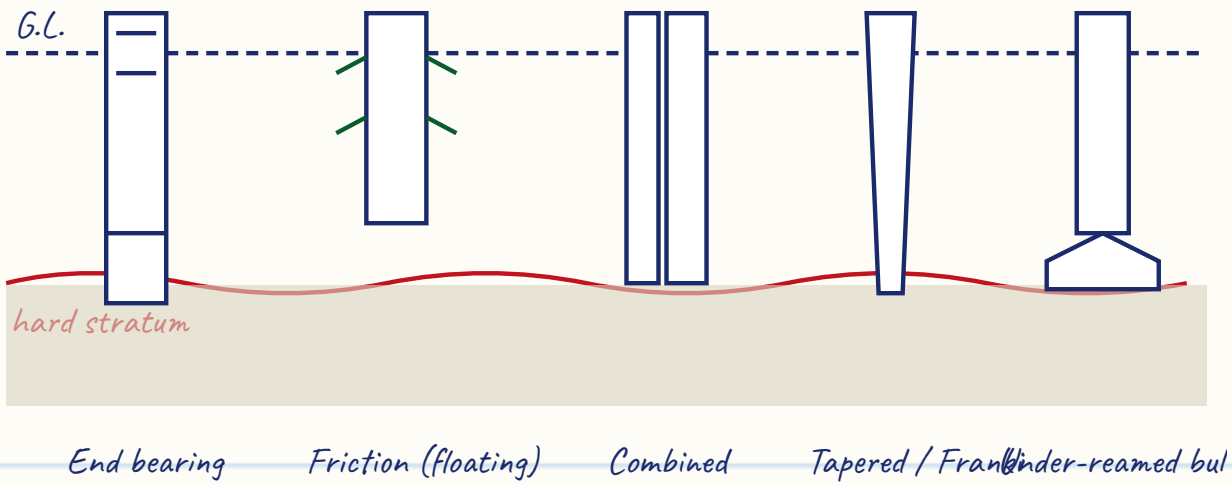


Fig 1: Pile types by load transfer & shape

Under-reamed pile (IS 2911 Pt 3): bulb dia = 2–3 × shaft; great in expansive (black cotton) soils.

5. Load Transfer Mechanism

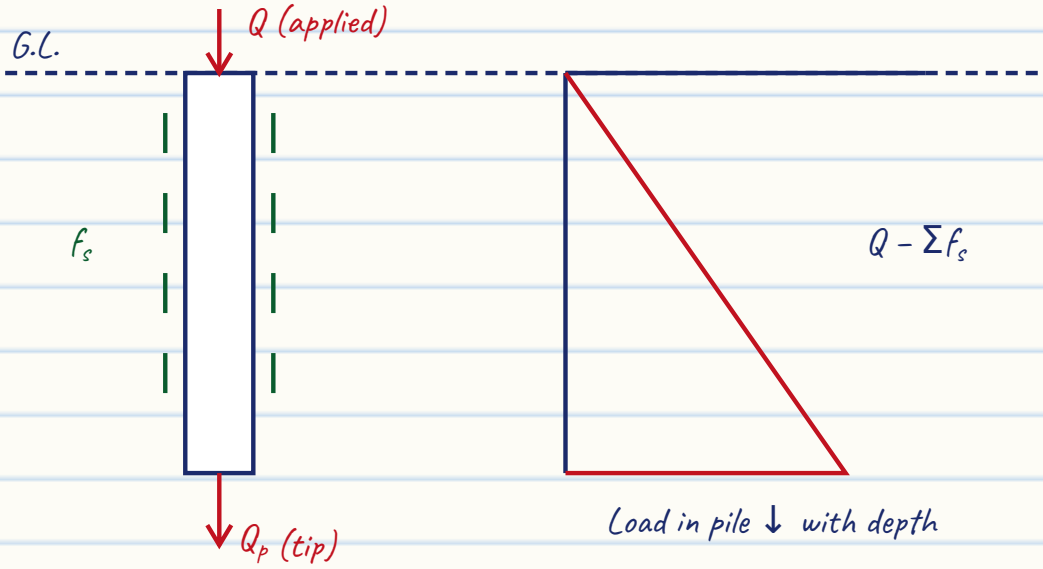


Fig 2: Shaft friction mobilises first (≈ 0.5–1 % d settlement); tip needs ≈ 10 % d

$$Q_u = Q_p + Q_s \quad ; \quad Q_a = Q_u / F.S.\ (2.5)$$

6. Load Capacity – Static Analysis

$$Q_u = q_p A_p + \Sigma f_s A_s - W_p(small)$$

A_p = tip area; A_s = perimeter × length; q_p = unit end bearing; f_s = unit skin friction.

(A) Piles in Clay ($\varphi = 0$; undrained)

End bearing:

$$Q_p = N_c\, c_u\, A_p \ , \ N_c = 9 \ (L/d > 4)$$

Skin friction – α method (total stress):

$$f_s = \alpha\, c_u \ \Rightarrow \ Q_s = \alpha\, c_u\, (\pi\, d\, L)$$

c_u (kPa)	≤ 25	25–50	50–75	> 75
α (typical)	1.0	0.8	0.6	0.4–0.5

- α = adhesion factor = adhesion / cohesion; ↓ as c_u ↑ (remoulding, water film)
- Layered clay: $Q_s = \Sigma\, \alpha_i\, c_{ui}\, P\, \Delta L_i$

λ method (Vijayvergiya & Focht) – driven piles in clay:

$$f_{av} = \lambda\, (\bar{\sigma}_v + 2\, c_u)$$

- $\bar{\sigma}_v$ = mean effective vertical stress along shaft; c_u = mean undrained strength
- λ depends on pile length (passive pressure effect):

L (m)	5	10	20	30	40
λ	0.34	0.25	0.17	0.14	0.13

β method (effective stress – NC clay long term):

$$f_s = \beta \sigma'_v, \beta = K \tan\delta \approx (1 - \sin\varphi') \tan\varphi' \approx 0.25 - 0.35$$

α → short-term (undrained). β → long-term / drained check. Use smaller of the two for design.

(B) Piles in Sand (c = 0 ; drained)

End bearing (Meyerhof):

$$q_p = \sigma'_v N_q \leq q_l = 0.5 P_a N_q \tan\varphi$$

- σ'_v = effective overburden at tip (limited to critical depth $z_c \approx 15\,d$ for loose–medium, 20 d dense sand)
- N_q from Meyerhof's chart (depends on φ and L/d; φ increased for driven piles: $\varphi = (\varphi_1 + 40)/2$)
- P_a = 100 kPa; limiting value q_l (the pile tip "critical depth" concept)

From SPT (Meyerhof):

$$q_p = 40\,\bar{N}(L_b/d) \leq 400\,\bar{N}\,(kPa)$$

$$f_s = 2\,\bar{N}(\text{driven}), \quad f_s = 1\,\bar{N}(\text{bored}) \quad (kPa)$$

Skin friction – β (effective stress) method:

$$f_s = K\sigma'_v \tan\delta \quad (\sigma'_v \text{ constant below } z_c)$$

Pile	K (loose → dense)	δ
Driven, displacement	1.0 – 2.0 ($K_0 = 1 - \sin\varphi$ to 1.75 K_0)	0.75 φ
Bored	0.7 – 1.0	2/3 φ
Steel H / pipe	0.5 – 1.0	20° (steel)

IS 2911 (Part 1/Sec 2) general formula:

$$Q_u = A_p(\tfrac{1}{2}\, d\gamma N_\gamma + \sigma'_v N_q) + \Sigma K\sigma'_{vi} \tan\delta\, A_{si}$$

Rock socketed pile:

- $q_p \approx 3\, q_{u,rock} \sqrt{(1 + 300\, t_d/s_d)}$ (Rowe-Armitage – brief) ; $f_s \approx 0.45 \sqrt{q_u}$ (MPa)

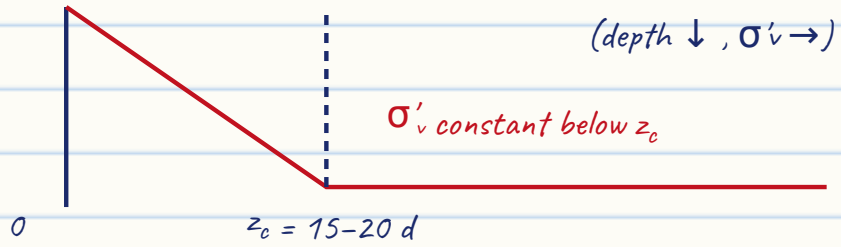


Fig 3: Critical depth – σ'_v and f_s cap

7. Worked Examples – Static Capacity

Example 1 (Clay)

A 400 mm dia. bored pile, 15 m long, in homogeneous clay c_u = 60 kPa, α = 0.6. Find safe load, F.S. = 2.5.

Solution:

$A_p = \pi/4 \times 0.4^2 = 0.1257\ m^2$; $A_s = \pi \times 0.4 \times 15 = 18.85\ m^2$

$Q_p = 9 \times 60 \times 0.1257 = \mathbf{67.9\ kN}$

$Q_s = \alpha\, c_u\, A_s = 0.6 \times 60 \times 18.85 = \mathbf{678.6\ kN}$

$Q_u = 67.9 + 678.6 = 746.5\ kN$

$$Q_a = 746.5 / 2.5 \approx 298.6\ kN \approx 300\ kN$$

Note: ≈ 91 % of capacity from skin friction → a friction pile.

Dynamic Pile Formulae

8. Dynamic Formulae

Idea: Energy delivered by hammer = work done in driving pile into soil + losses. Gives capacity at the time of driving – quick field check for driven piles in granular soils (unreliable in clay: set-up / relaxation).

$$W H \eta_h = Q_u \times S + \text{losses}$$

(i) Engineering News Formula (ENR)

- Drop hammer: $Q_u = W H / (S + 2.5)$ (S in cm, WH in kN·cm), F.S. = 6
- Single-acting steam hammer: C = 2.5 cm ; double-acting: C = 0.25 cm

$$Q_a = W H / [6 (S + C)]$$

W = weight of hammer (kN), H = height of fall (cm), S = final set per blow (cm), C = 2.5 cm (drop).

(ii) Hiley Formula (considers hammer efficiency, restitution, temporary compression)

$$Q_u = [\eta_h W H / (S + C/2)] \cdot [(W + e^2 P) / (W + P)]$$

- η_h = hammer efficiency (0.75–0.85 drop; 0.8–0.9 steam; 0.8 diesel)
- P = weight of pile + helmet + cap ; W = hammer weight
- e = coefficient of restitution : 0.25 (wood), 0.4 (conc. with cushion), 0.5 (steel on steel)
- $C = C_1 + C_2 + C_3$ = temporary elastic compression of (cap + pile + soil)
- Typical: C_1 (cap) 5–10 mm, C_2 (pile) 2–5 mm, C_3 (soil, quake) 0–5 mm … F.S. = 2–3

(iii) Modified Hiley (IS 2911 Part 1) – uses efficiency of blow:

$$Q_u = \eta_h \cdot \eta_b \cdot W H / (S + C/2)$$

$\eta_b = (W + e^2 P) / (W + P)$ = efficiency of blow; if $P > W e^2$, η_b falls → heavy pile needs heavy hammer.

(iv) Danish Formula

$$Q_u = \eta_h E / [S + \sqrt{(\eta_h E L / 2 A E_p)}]$$

- $E = W H$ (hammer energy), L = pile length, A = cross-section area, E_p = Young's modulus of pile
- Second term in denominator = elastic compression of pile (accounts for pile length) ; F.S. = 3

Example 3 (Dynamic formulae)

A precast RC pile (0.3 m × 0.3 m, 12 m, 25 kN/m^3 ⇒ weight P = 27 kN) is driven by a drop hammer of W = 25 kN, fall H = 2.5 m. Average set = 15 mm / blow. Find (a) allowable load by ENR; (b) Hiley: η_h = 0.8, e = 0.4, C = 12 mm (F.S. 2.5).

(a) ENR (drop hammer)

$W H$ = 25 × 250 cm = 6250 kN·cm ; S = 1.5 cm ; C = 2.5 cm

$$Q_a = 6250 / [6 (1.5 + 2.5)] = 6250 / 24 \approx 260 \text{ kN}$$

(b) Hiley

$\eta_b = (W + e^2 P)/(W + P) = (25 + 0.16 \times 27)/(25 + 27) = 29.32 / 52 = 0.564$

$W H \eta_h$ = 0.8 × 25 × 2.5 = 50 kN·m = 50 000 kN·mm (units of S in mm)

$S + C/2$ = 15 + 6 = 21 mm

$Q_u = (50\,000 / 21) \times 0.564 = 2381 \times 0.564 = 1343$ kN

$$Q_a = 1343 / 2.5 \approx 537 \text{ kN}$$

ENR is very conservative (F.S. 6) – gives lower value; Hiley is closer to the load-test results.

9. Pile Load Test (IS 2911 Part 4)

Direct, most reliable method – conducted on **initial test pile** (to establish capacity, test to 2 × design load) and **routine test pile** (on working pile, 1.5 × design load).

Types

1. **Maintained Load Test (MLT / ML)** – load applied in increments; settlement observed
2. **Constant Rate of Penetration (CRP)** – pile pushed at fixed rate; load recorded
3. **Cyclic Load Test** – loading–unloading cycles to separate elastic & plastic settlement

Set-up

- **Kentledge** (dead weight ≥ 1.25× max. test load on a platform) or **reaction piles** with anchored beam (≥ 3 d clear spacing)
- Hydraulic jack + calibrated pressure gauge/load cell; dial gauges (0.01 mm) on independent reference beam

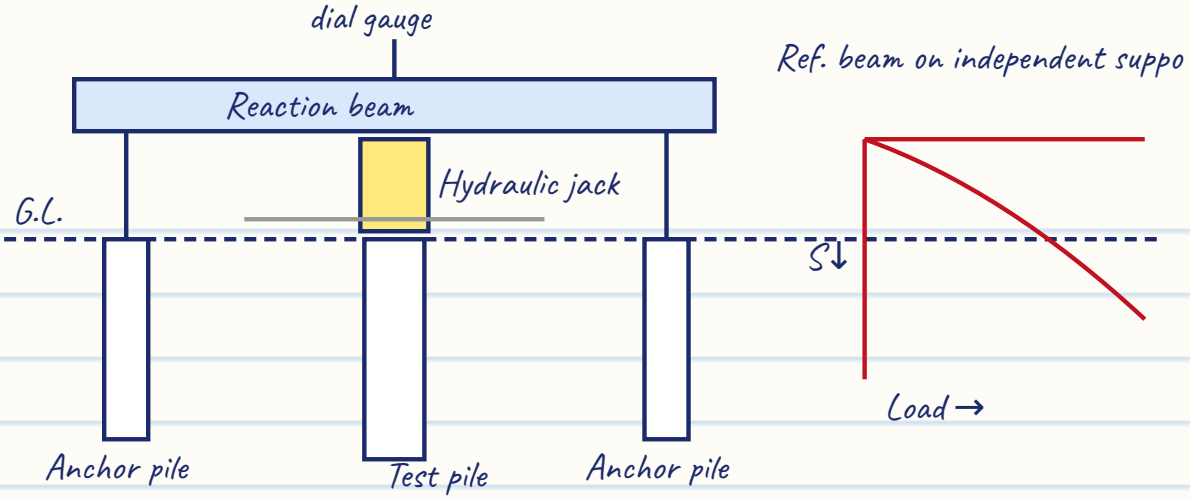


Fig 4: Pile load test by reaction piles; typical load–settlement curve (right)

Maintained load test – procedure

- Apply load in **equal increments = 20 % of safe load**; hold each until rate of settlement ≤ 0.1 mm/h (min. 1 h; or 2 h)

- Continue to 2 × design load (or failure); hold final load 24 h; then unload in same stages
- Read settlement at 5, 10, 20, 30, 60 min … each stage

CRP test

- Jack forces pile at constant rate: **0.75 mm/min (sand), 0.25 mm/min (clay)**; load measured continuously
- Ends at ultimate load or (settlement = 10–15 % d); quicker – hours not days

Interpretation – allowable load (IS 2911 Pt 4) = least of

1. ⅔ of load causing **12 mm** total settlement
2. ½ of load causing **10 % of pile dia.** settlement
3. ⅔ of load for 6 mm net (after rebound) settlement (some cases)

Plot load–settlement; ultimate load by tangent/ double-tangent intersection or Davisson offset (0.15 + d/120 in.).

Pile Groups

10. Pile Groups

Piles usually used in **groups** (3–9 piles) connected by a **pile cap**. Capacity of group ≠ N × single pile because stress zones overlap.

$$\text{Group efficiency } \eta_g = Q_{g(u)} / (n\,Q_u)$$

Spacing

- Min. c/c = 2.5 d (IS: 2.5 d end-bearing; 3 d friction). Typical 3 d
- Edge distance ≥ 1.5 d from pile centre to cap edge

Piles in Sand

- Driven piles densify sand → η_g may be > 1 (spacing 3–4 d). For bored piles $\eta \approx$ 2/3–1.
- Design: $Q_g = n\,Q_u$ (η = 1, for $s \geq 3d$) ; settlement governs

Piles in Clay

- Capacity = **lesser** of (i) $n \times Q_u \times \eta$ and (ii) **block failure** of soil + piles acting as an equivalent pier

$$Q_{g(u)} = 2\,(L_g + B_g)\,D\,\bar{c}_u + N_c\,c_u\,L_g\,B_g$$

L_g, B_g = overall plan dimensions of group (out-to-out) ; D = pile length ; N_c from Skempton ($\approx$ 9, $5(1+0.2B/L)(1+0.2D/B)$).

Converse–Labarre formula (n_1 rows × n_2 columns)

$$\eta_g = 1 - \theta\,[(n_1-1)\,n_2 + (n_2-1)\,n_1] / (90\,n_1\,n_2)$$

$\theta = \tan^{-1}(d/s)$ in degrees; d = pile diameter, s = c/c spacing.

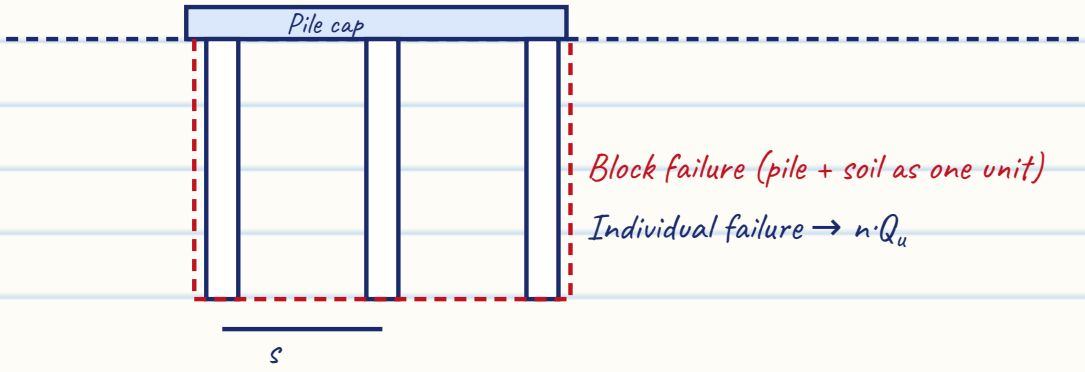


Fig 5: Block failure of pile group in clay

Example 4 – Group capacity

A 3 × 3 group of 400 mm dia. piles, 12 m long, c/c spacing 1.2 m, in clay c_u = 50 kPa, α = 0.7. Find group capacity (F.S. = 2.5).

Single pile: $Q_p = 9 \times 50 \times 0.1257 = 56.5$; $Q_s = 0.7 \times 50 \times \pi \times 0.4 \times 12 = 527.8$; Q_u = **584 kN**

(i) Individual action (Converse–Labarre):

$\theta = \tan^{-1}(0.4/1.2) = 18.43°$; $\eta = 1 - (18.43/90)(2\times3 + 2\times3)/9 = 1 - 0.2048 \times 1.333$ = **0.727**

$Q_g = 0.727 \times 9 \times 584 =$ **3821 kN**

(ii) Block failure: $B_g = L_g = 2 \times 1.2 + 0.4 = 2.8$ m

Perimeter shear = $4 \times 2.8 \times 12 \times 50 = 6720$ kN ; base = $9 \times 50 \times 2.8^2 = 3528$ kN

Q_g = 10 248 kN ($\gg$ 3821)

Governing $Q_{g(u)}$ = 3821 kN $\Rightarrow$ Q_a = 3821/2.5 $\approx$ 1528 kN

II. Settlement of Pile Groups

- Single pile settlement small; group settles more – deeper stress bulb ($\approx$ 1.5 B_g)
- **Equivalent raft method (clay)** : load acts on imaginary raft at **2L/3** (friction piles) or at pile tip (end-bearing) ; spread by **2 V : 1 H** ; compute consolidation settlement

$$S_g = \Sigma\,[C_c\,H / (1+e_0)]\,\log[(\sigma'_0 + \Delta\sigma')/\sigma'_0]$$

- **Sand (Vesic):** $S_g = S\sqrt{(B_g / d)}$ (S = single pile settlement)
- **Meyerhof (SPT):** S_g (mm) $= 0.96\, q\sqrt{(B_g I_c)} / N$; $I_c = 1 - L/(8B_g) \geq 0.5$
- Permissible: isolated 12–25 mm (IS); differential 1 in 500.

12. Negative Skin Friction (Down-drag)

Soft layer / fill settling faster than the pile ⇒ soil drags pile **downward** – extra load!

- **Causes:** fill on soft clay, lowering of water table, consolidating clay, collapse of soil
- Neutral plane: level where soil & pile settle equally

$$Q_n = P \cdot L_n \cdot f_n ; f_n = c_u \text{ (clay) or } \beta\, \sigma'_v \text{ (sand)}$$

- Design: $Q_a = (Q_u - Q_n) / F.S.$ (friction piles) ; for end-bearing: pile structural load = $Q + Q_n$
- **Reduce:** bitumen coating, sleeves, larger preload, increase pile dia. (less area/perimeter)

Well Foundations

13. Well (Caisson) Foundation – Introduction

A **well foundation** is a hollow masonry / RC box (circular, rectangular ...) sunk by excavating soil inside, then plugged and filled – used for **bridge piers & abutments** in rivers; depth 20–40 m or more. India: typical for large rivers (Ganga, Yamuna, Godavari).

Disadvantages

- Tilts & shifts; difficult to sink through boulders / clay (hang-up); needs skilled supervision

14. Types of Well (Caisson) Foundations

1. **Open caisson (open well)** – open at top & bottom; sunk by dredging through the open; most common for bridges. Variants: Well/block, pier-well, box.
2. **Box caisson** – closed at bottom, open at top; prefabricated on land, floated, sunk on prepared bed (levelled, firm soil, water depth ≤ 12 m; docks, jetties)
3. **Pneumatic caisson** – closed top; **compressed air** keeps water out in working chamber; for very deep (> 30 m) bed, no dredging; caisson disease for workers; costly; max. ~ 35 m (3.5 atm)
4. **Floating caisson, monolith** (multi-dredge-well)

Type	Open	Box	Pneumatic
Bottom	Open	Closed	Working chamber
Top	Open	Open	Closed (air lock)
Depth	Large (≤ 40 m)	≤ 12 m	≤ 35 m
Use	Bridge piers	Harbours	Deep, clay/boulders

15. Shapes of Wells

- **Circular** – most common: uniform stresses, least perimeter, easy sinking, minimum tilt, good for river currents from any angle
- **Double-D** – oblong piers, two dredge holes; **Twin circular** – wide piers/ abutments
- **Double octagonal** – 2 dredge holes (aligned with flow)
- **Rectangular / Square** – large, multiple dredge holes; need corner stiffening

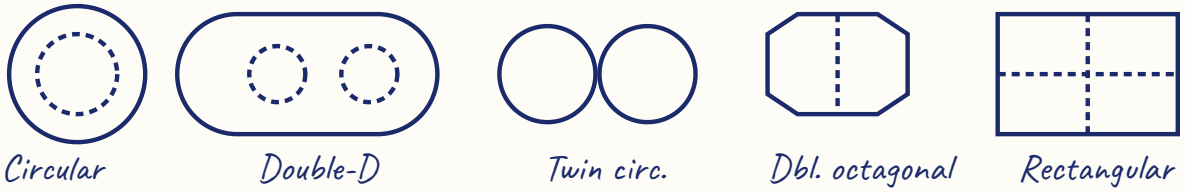


Fig 6: Plan shapes (dotted = dredge holes)

16. Components of a Well Foundation

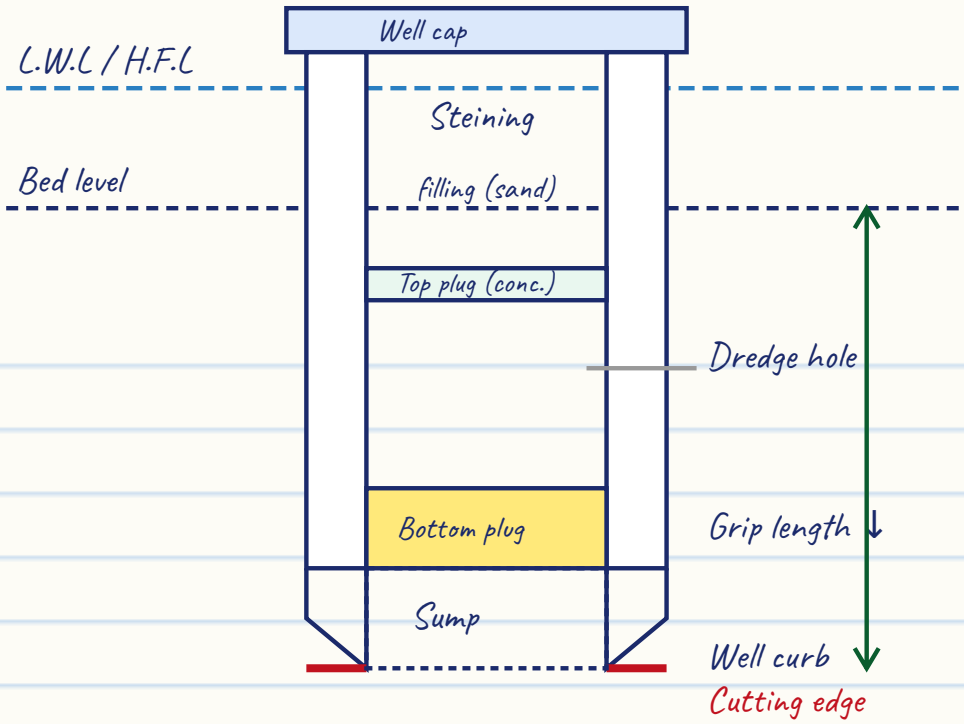


Fig 7: Section through an open-type well

Component	Function / Detail
Cutting edge	Steel angle / plates at bottom; cuts soil; MS, 20 mm+ thick
Well curb	RC wedge above cutting edge, inner face 30°–37° to horizontal; strong; M25 conc.
Steining	Well wall; brick / stone masonry or RCC; min. 500 mm; heavy enough to sink & resist skin friction
Bottom plug	Concrete (tremie) filling under water, 0.6–1 m above curb top ; forms the seal
Sand filling	Fills the well between plugs – economical; gives weight
Top plug	≥ 300 mm plain concrete under well cap
Well cap	RC slab at top; distributes pier loads ; ≥ 750 mm thick
Sump	Extra excavation below cutting edge ≥ 0.5 m (1.5 m max) ; allows seating

17. Functions of Well Foundations

- Transfer large vertical loads (via base + skin friction) to sound strata
- Resist lateral loads (passive earth pressure on sides) & overturning
- Provide grip length below scour level – safeguard against scour
- Act as a cofferdam / retaining structure to excavate in water
- Resist impact of floating debris, boats; give mass for pier / abutment

18. Forces on Well Foundations (IRC 6 / 78)

Force	Details
Dead load	Superstructure + pier + cap + well (buoyancy reduces)
Live load + impact	Vehicular load, impact factor (IRC Class A/70R)
Wind	On superstructure, live load, exposed pier
Water current	$P = 52\,K\,V^2$ (kg/m^2), K = 1.0 square, 0.66 circular piers; V = velocity (m/s) at the max scour...(varies with depth)
Tractive / braking	Longitudinal horizontal force
Earth pressure	For abutment wells: Rankine / Coulomb
Buoyancy	Upward = weight of displaced water (full for soil less than 1m...)
Seismic, temperature, shrinkage, centrifugal	As per IRC 6

Scour (Lacey)

$$d_{sm} = 0.473\,(Q/f)^{1/3} \;;\; \text{max. scour} = 2\,d_{sm} \text{ (at pier nose)}$$

- f = Lacey silt factor = $1.76\sqrt{d_{mm}}$; Q = design discharge (m^3/s)
- Founding level = max. scour + **grip length** (≥ 1/3 × max scour depth below M.S.L.); in rock ≥ 0.6 m socket

Stability checks

- Base pressure ≤ safe bearing capacity ($\sigma_{max} = P/A + M/Z$)
- Lateral stability – **Terzaghi / IRC** method: passive resistance of soil mobilised on side faces; $\sigma \le$ allowable passive pressure ÷ F.S. (2)
- Tension in steining/ well cap; check in-flow & sliding

Max/min base pressure: tension never allowed at base (kern); IRC permits 25 % more stress with wind/seismic.

19. Sinking of Wells

Well sunk under its own weight by **excavating soil from inside** (dredge hole); steining is built in **lifts of 1.5–2 m**, each allowed to set before sinking.

1. **Preparation:** in water → build **sand island** / cofferdam / floating caisson; on land → level bed
2. **Laying the cutting edge & curb** on prepared platform; centre accurately; build first lift of steining
3. **Dredging** with **grab (clamshell)**, dredge-hole crane or divers; evenly from all dredge holes
4. Well sinks when W_{well} + kentledge > skin friction + end resistance (check each stage)
5. Repeat lifts, check **tilt (plumb)** and **shift** each shift
6. At founding level: inspect strata, clean, place **bottom plug** by tremie, fill, **top plug**, cap

Aids to sinking

- **Kentledge** (extra load – rails, sand bags, concrete blocks)
- **Water jetting** (outside) / **bentonite lubrication** – reduces skin friction
- **Blasting** (minor, with caution) for hard strata / boulders ; air-jets
- Dewatering (bailing) to allow dry dredging (reduce buoyancy)
- Pneumatic cavity for obstruction removal

Difficulties

- **Hang-up** – friction ≫ weight; **sudden sinking / blow** in quicksand; obstructions (tree trunks, boulders); skin friction ↑ in stiff clay

20. Tilts and Shifts

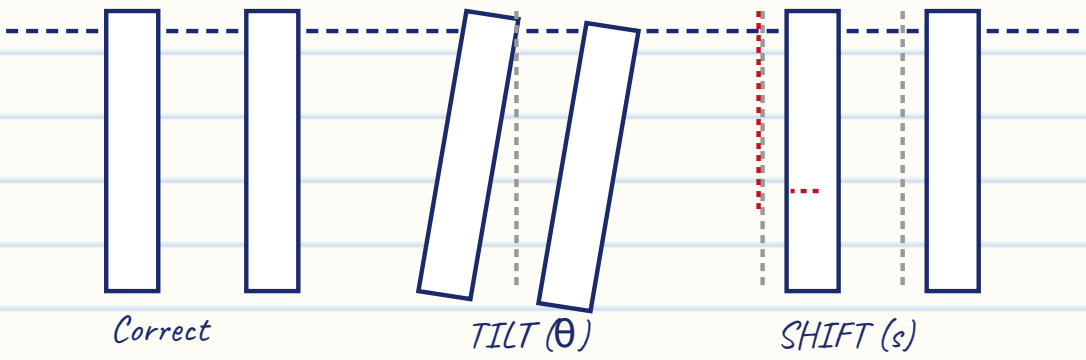


Fig 8: Tilt = angular deviation from vertical; Shift = lateral displacement of centre

- **Tilt** = H : V deviation (e.g. 1 in 100) ; **Shift** = horizontal displacement of centre at base. IRC limits: tilt ≤ 1 in 80 (during sinking: 1 in 100), shift ≤ 150 mm

Causes of Tilts & Shifts

- **Uneven strata** – hard patch / sloping rock on one side, boulders, tree trunks
- Unequal dredging among dredge holes; sudden collapse of soil / blow
- Eccentric kentledge; non-uniform skin friction; unequal steining weight
- Scour on one side; current / wind pressure; imperfect first lift (curb not horizontal)
- Sudden release of friction while sinking

Remedial measures – Tilt

- **Eccentric dredging** – dredge more from the high side (side opposite to tilt direction), leave soil on low side
- **Eccentric loading** (kentledge) on the high side
- **Water jetting / pulling** with wire ropes + winches (anchored); strutting on low side with piles/ sandbags
- **Obstruction removal** by divers, blasting or chiselling (chisel bar)
- Reduce skin friction on the high side; dewatering

Remedial measures – Shift

- Apply **horizontal pull** with ropes, tackle; **strutting** / anchor against the shifted side
- Induce tilt deliberately opposite then correct; dredge / water jet on the side to which the well should move

- **Prevention:** accurate setting, sand island level, **guide frames**, check every 1 m sink with plumb bob / theodolite

Tilt (1 in n) = (shift of base centre – shift at top) / depth sunk

Total shift ≈ tilt × depth + original shift. Tilt corrected gradually – never rush.

Important Questions (Unit V)

1. Classify piles; explain load-carrying mechanism with sketches. (Types: friction/end-bearing)
2. Derive load capacity of a pile in clay (α method) and sand. **(Num.)**
3. Explain Hiley's formula; compare with ENR & Danish. Mention the limitations of dynamic formulae.
4. Describe the pile load test; how is the safe load determined? CRP v/s maintained load test.
5. What is group efficiency? Converse–Labarre; block failure of pile group in clay. **(Num.)**
6. Settlement of a pile group; equivalent raft method. Negative skin friction – causes & remedies.
7. Draw a neat sketch of a well foundation, label components, functions of each.
8. Types and shapes of wells; forces acting; how is a well sunk?
9. Causes of tilts & shifts and remedies – short note.
10. Pneumatic caisson vs open caisson; scour depth & grip length.